

Final Project

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2026-04-09

Problem Definition

A surplus production model is a simple population dynamics model used for fisheries stock assessment. The objective of a surplus production model is to estimate the Maximum Sustainable Yield (MSY) using limited data, to identify the optimum harvest effort that avoids depleting the stock. The equation for the surplus production model is:

$$B_{t+1} = B_t + rB_t(1 - B_t/K) - qEB_t$$

Where B is the stock biomass at time t r is the intrinsic growth rate K is the environmental carrying capacity (units/cell) q is the catchability coefficient E is fishing effort

The catchability coefficient (q) relates fishing effort to fishing mortality and is assumed to be constant in this equation.

However, the true value of the catchability coefficient changes with increased fishing efficiency, fish density (hyperstability or hyperdepletion), environmental changes, or technological improvements. This has the potential to impact the determined MSY of a stock, having implications for policy development and fisher economic gain. As such, my project aims to evaluate whether assuming a constant catchability coefficient (q) in a Bayesian stock assessment leads to biased estimates when fishers are adaptively targeting aggregations.

The POSEIDON model is an interactive agent-based modeling platform that serves as an interactive tool that encodes certain parameters within a dynamic spatial-visual platform. Users of the interface apply values to these parameters to evaluate the effect of various policy interventions on ecological and fisheries outcomes. Using this model, I will be able to simulate fisheries data where fishers are highly exploratory versus where fishers use explore-exploit-imitate to aggregate effort on high-density patches through patterns of exploitation and imitation.

With these two datasets, I will run six models – fitting Bayesian surplus production models with both uninformative and informative priors (assuming good biological data for the simulated fish species) to evaluate:

- 1) How does adaptive fisher behavior bias posterior estimates of MSY in a Bayesian surplus production model that assumes constant catchability?
- 2) How does this misspecification affect posterior uncertainty, including CI width?
- 3) To what extent can informative priors mitigate bias arising from misspecified catchability?
- 4) Do Bayesian models that allow for flexible or time-varying catchability better recover true MSY under adaptive fishing behavior?

Simulation of Data

The POSEIDON model was used to simulate fishing scenerios with two different types of fisher behavior. These fisher simulation runs differ only in fisher targeting behavior, all else being held constant.

In Scenerio A - fishers move randomly, catch is roughly proportional to abundance. This means that CPUE is a good index of biomass and constant q is approximately valid.

In Scenerio B - fishers use POSEIDON's adaptive explore-exploit-imitate algorithm to seek out high-density patches. Agents learn to efficiently target remaining fish aggregations, depleting nearby areas first and expanding outward. CPUE stays high as biomass drops (hyperstability), meaning the constant q assumption is violated.

The constant parameters of both simulations are:

$K = 3000$ (max fish per cell) * 2500 cells_sq $K = 7500000$ fish in the system

$r = 0.35$

$q = 0.002$

These parameters result in a $MSY_{true} = 656,250$

Fleet size = 100

Speed = 5 km/h

Maximum of 7 days at sea

12 hours between each departure

Fuel tank size = 100,000 liters

Liters/km = 10

Gas price/liter = 0.01

Market price of fish = \$10

Hold size = 200

Each fisher has two information sharing 'friendships'

Scenerio A - Fixed behavioral probability

Exploration probability = 0.95 (high)

Imitation probability = 0.0 (no imitation)

Scenerio B - Adaptive behavioral probability

Exploration probability = 0.05 (low random exploration) Min = 0.01

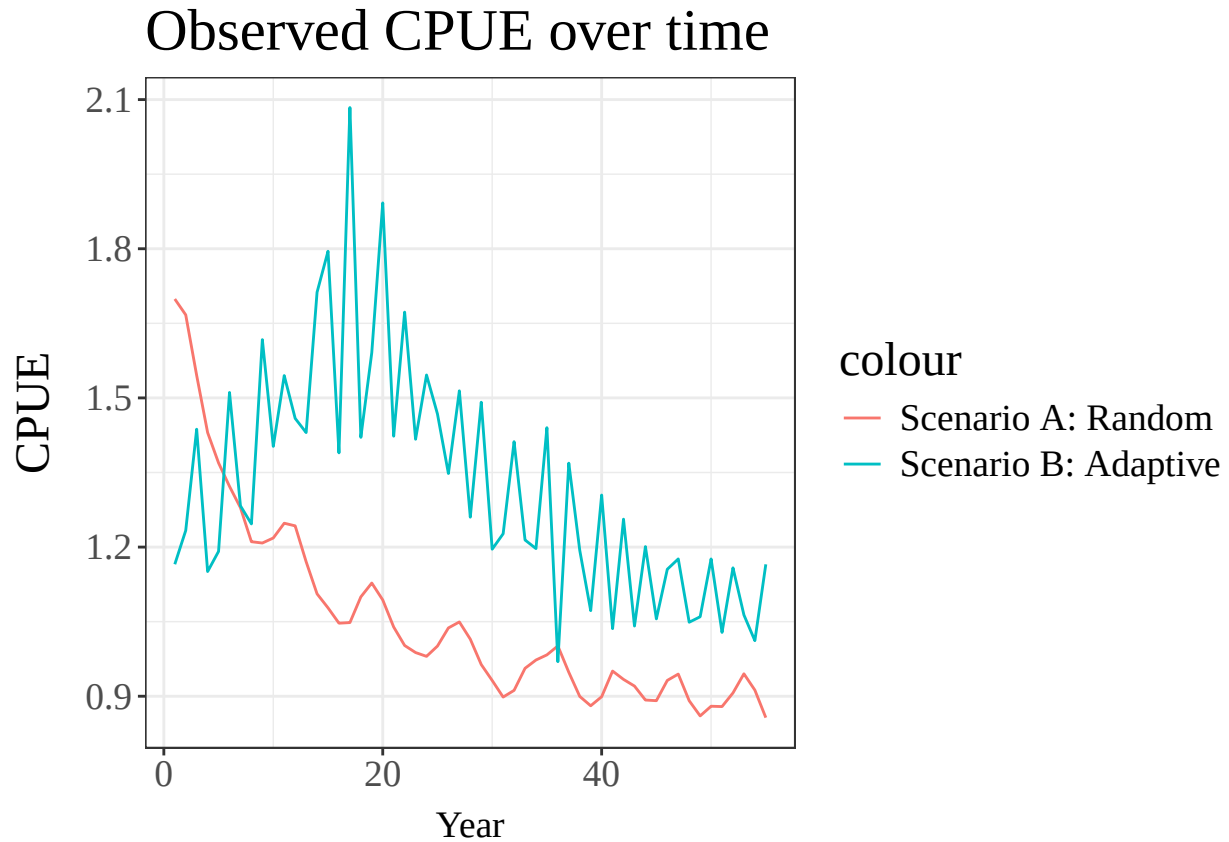
Imitation probability = 1.0 (high social imitation) Increment multiplier= 0.03 (learning speed)

Dataset

A simulation of 30 years was run for each scenario with the given parameters.

Species biomass (kg), total catch (kg), average distance from port (km), average last trip hourly income (\$), hours out, and effort were collected from the simulation. CPUE was calculated from total catch (kg) and effort.

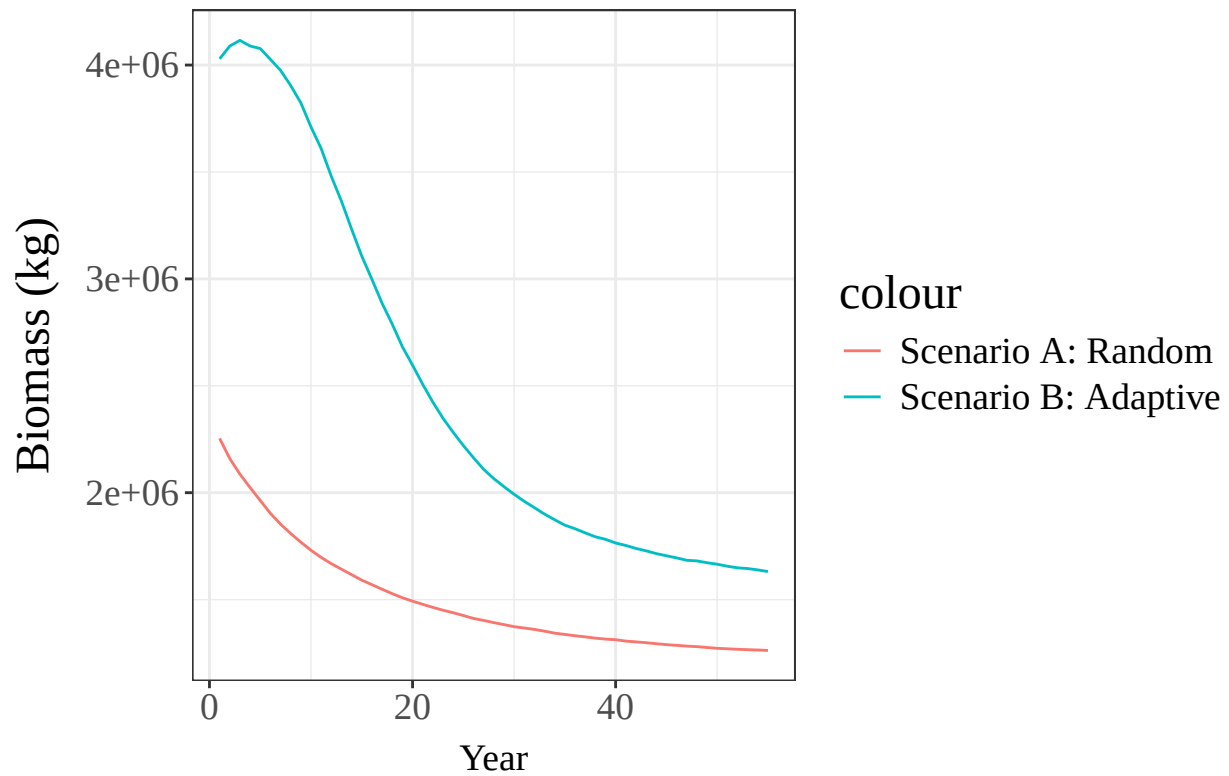
Two runs with random seeds were conducted for each scenario. The first 5 years were removed to drop potential trends arising from initialization and to remove the learning artifact in the Scenario B.



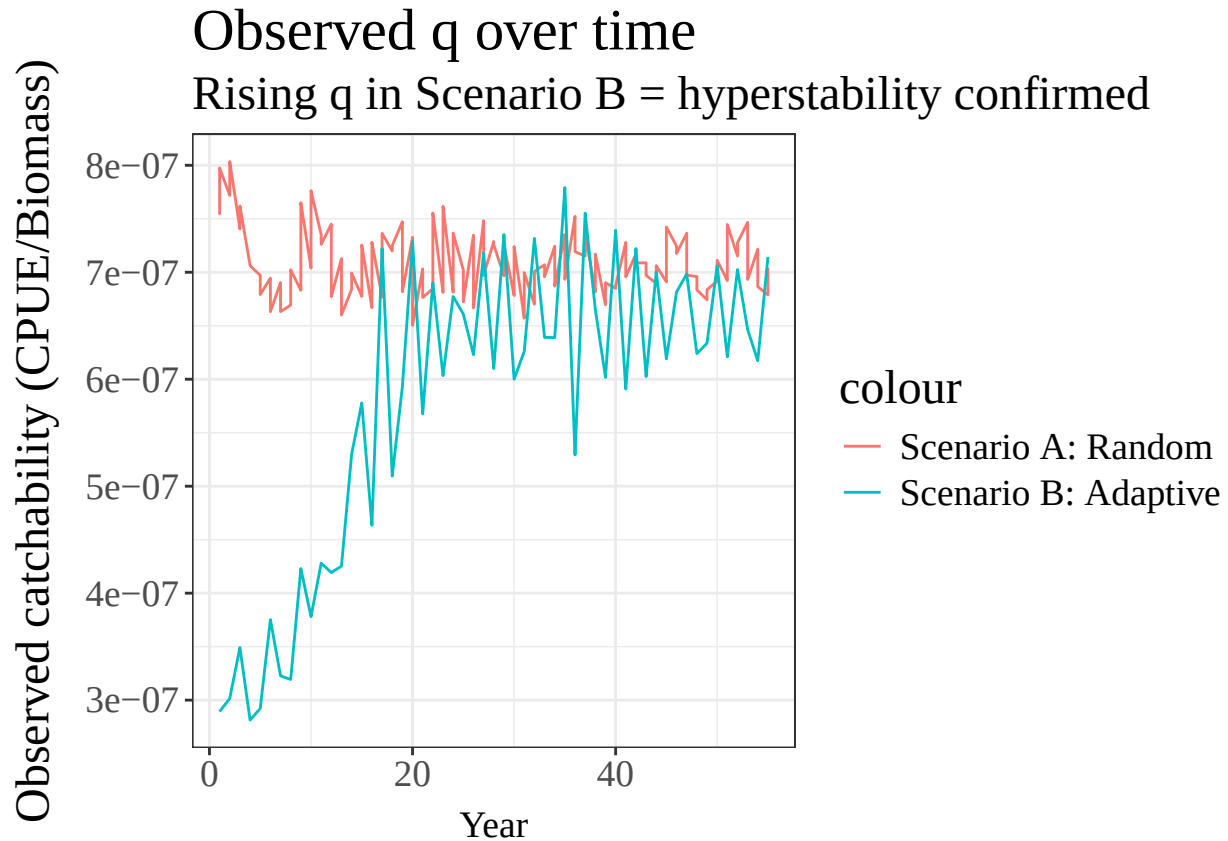
Scenario A shows a steady decrease in CPUE over time. As a result of highly exploratory behavior, the decrease in biomass (seen below) is directly tied to a decrease in CPUE.

Scenario B shows an initial increase in CPUE given higher biomass and then declines more slowly than in Scenario A. Adaptive behavior enables fishers to imitate and exploit centers of high biomass, leading to increased instability in CPUE as fishers find and deplete these centers.

Observed Biomass over time



Both Scenarios at year 0 have the same initial biomass. In Scenario A, biomass rapidly decreases initially and steadies to a point of limited decrease in later years, but remains above zero. In Scenario B, the initial biomass spike is caused by the inefficiency associated with learning. However, Scenario B shows a similar pattern of rapid decline followed by stabilization.



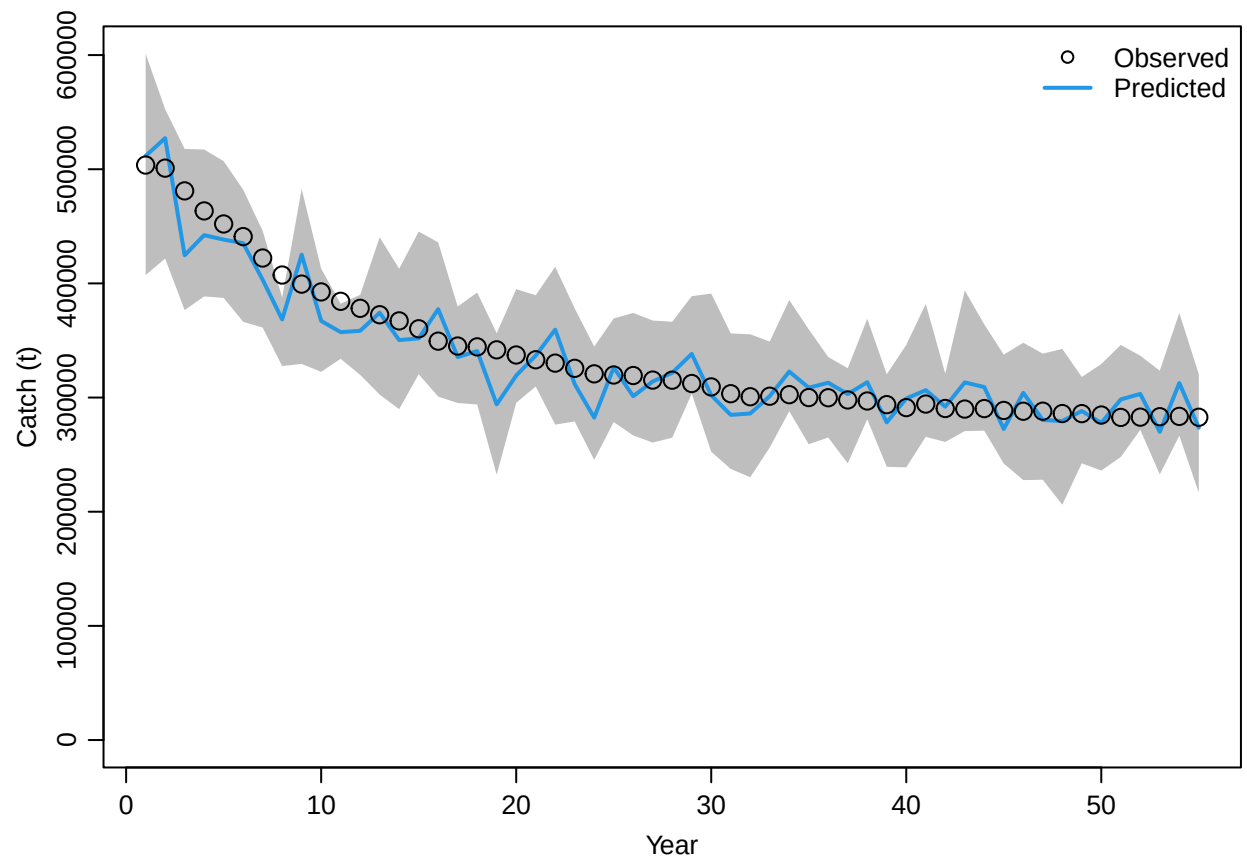
Scenario A represents consistency with the assumption for a standard population surplus model. The relationship between CPUE and biomass creates a largely flat trend where CPUE decline is proportional with biomass decline over time. The trend indicates that a constant q assumption is largely valid.

In contrast, Scenario B shows hyperstability. The relationship between CPUE and biomass is initially low due to learning restrictions, then increases despite decreasing biomass as a result of advantageous exploitation and imitation behavior. Spatial targeting results in a more volatile pattern as established previously. The trend indicates that a constant q assumption is not valid.

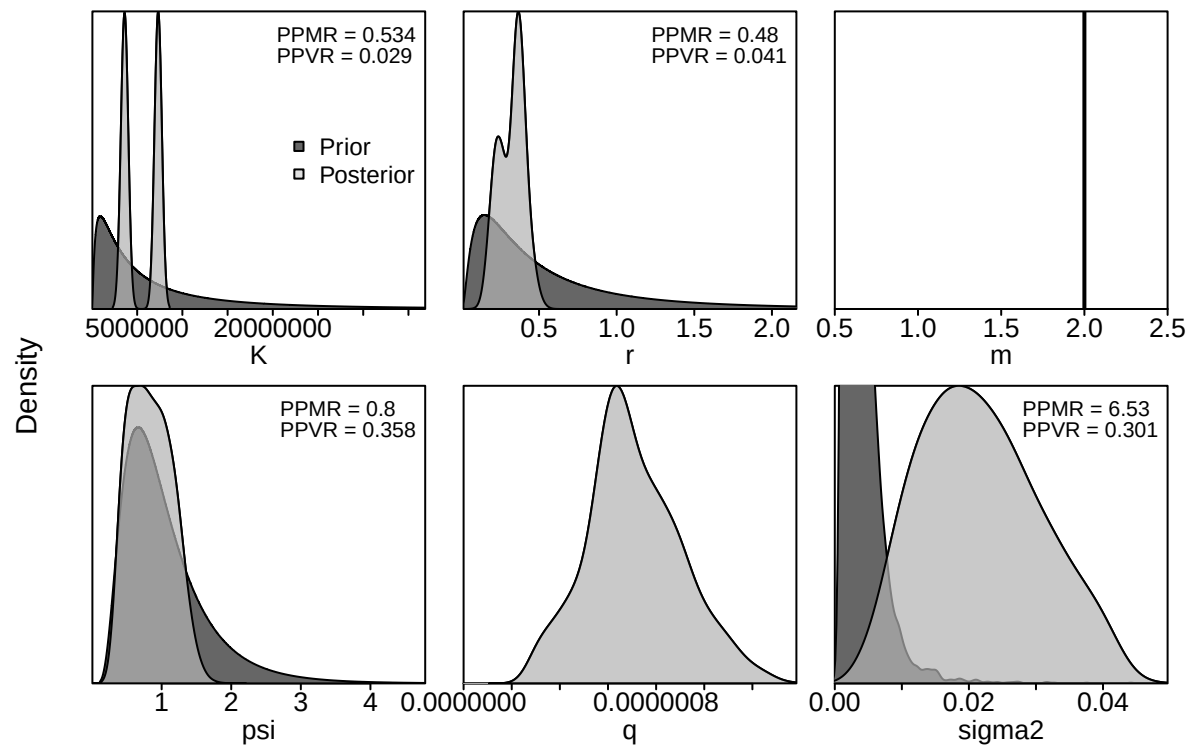
Model 1: Standard Surplus Production

Scenario A

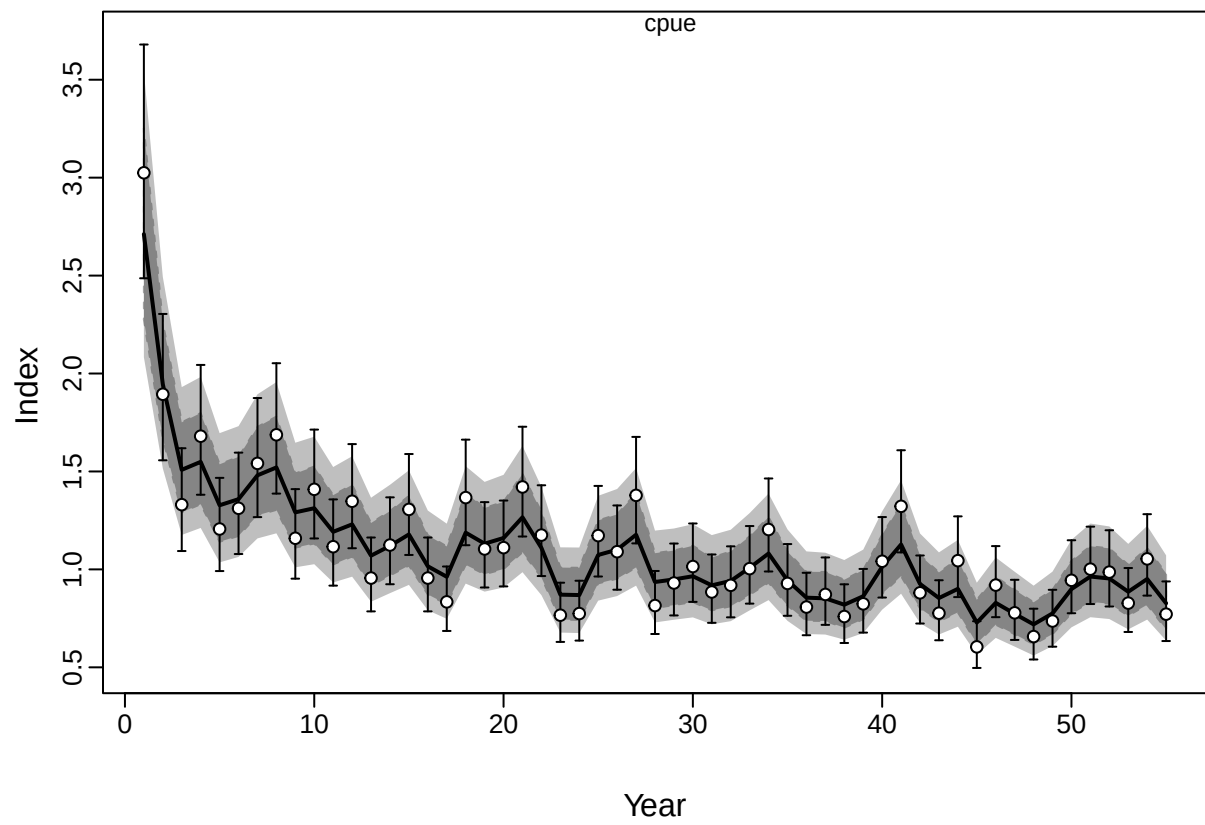
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##
## ><> jplot_catcherror() <><
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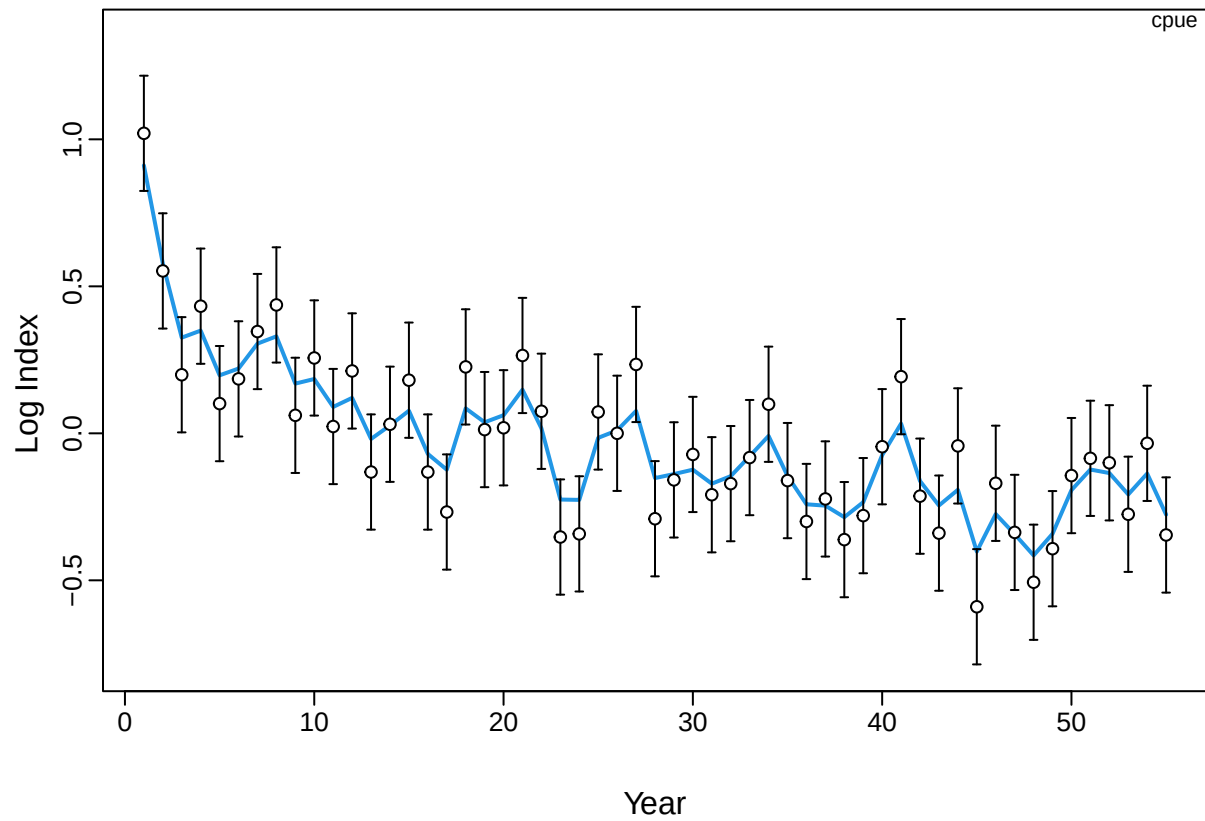
```
##
## ><> jbplot_ppist() - prior and posterior distributions <><
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##
## ><> jbplot_cpue() - fits to CPUE <><
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## ><> jplot_cpue() - fits to CPUE <><
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##
## ><> jplot_trj() - B trajectory <><

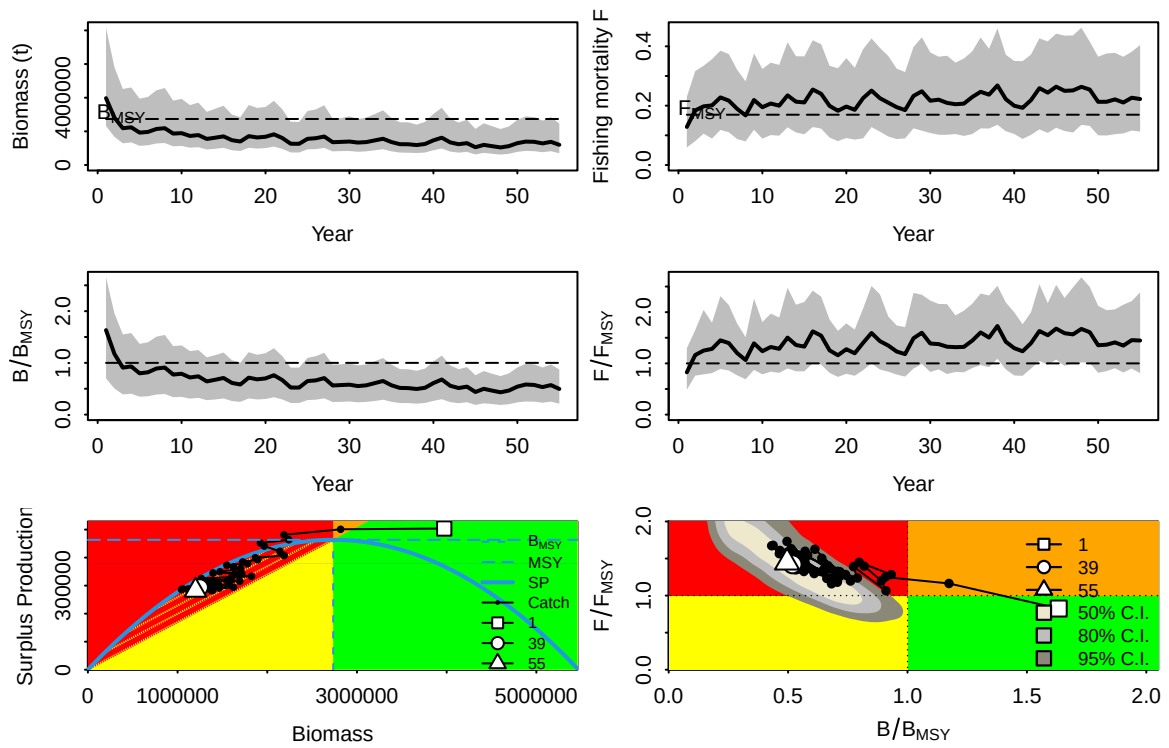
##
## ><> jplot_trj() - F trajectory <><

##
## ><> jplot_trj() - BBmsy trajectory <><

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## ><> jplot_trj() - FFmsy trajectory <><

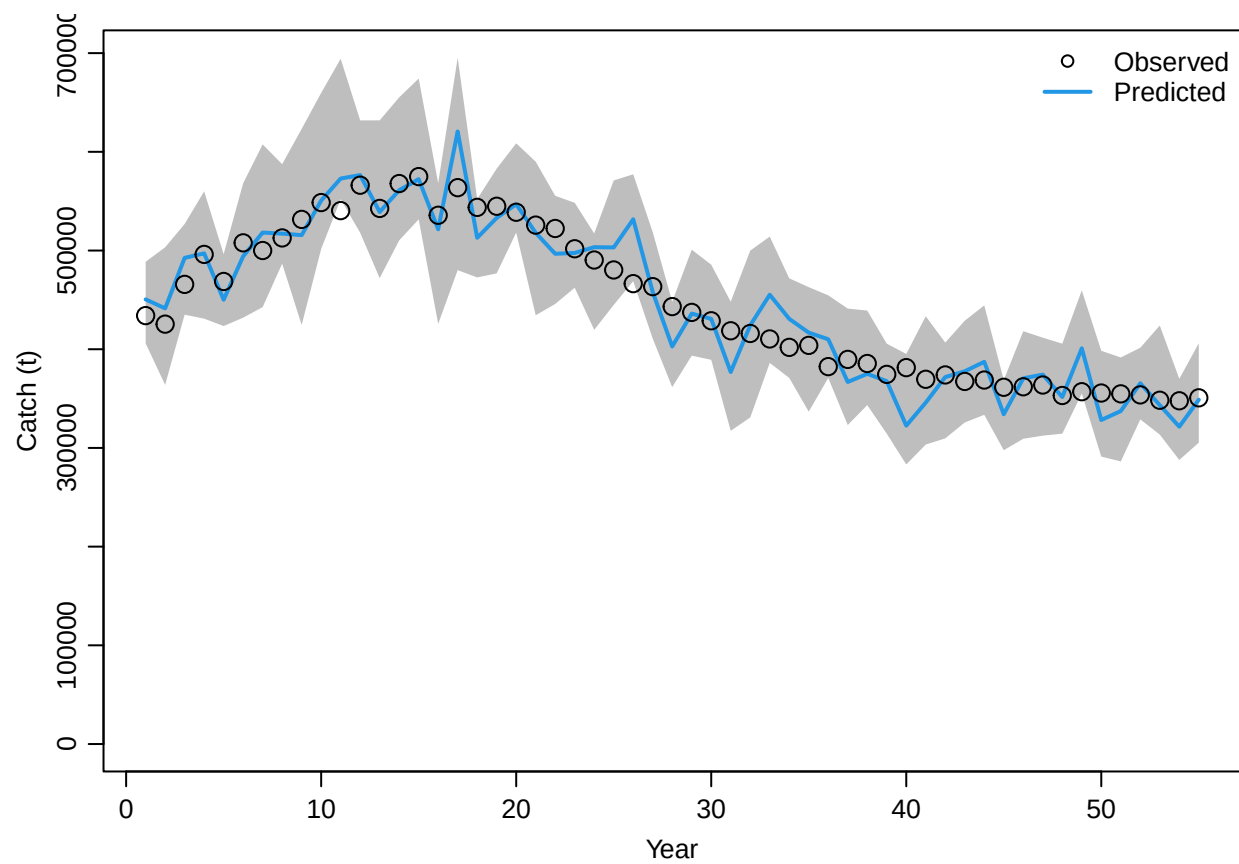
##
## ><> jplot_spphase() - JABBA Surplus Production Phase Plot <><

##
## ><> jplot_kobe() - Stock Status Plot <><
```

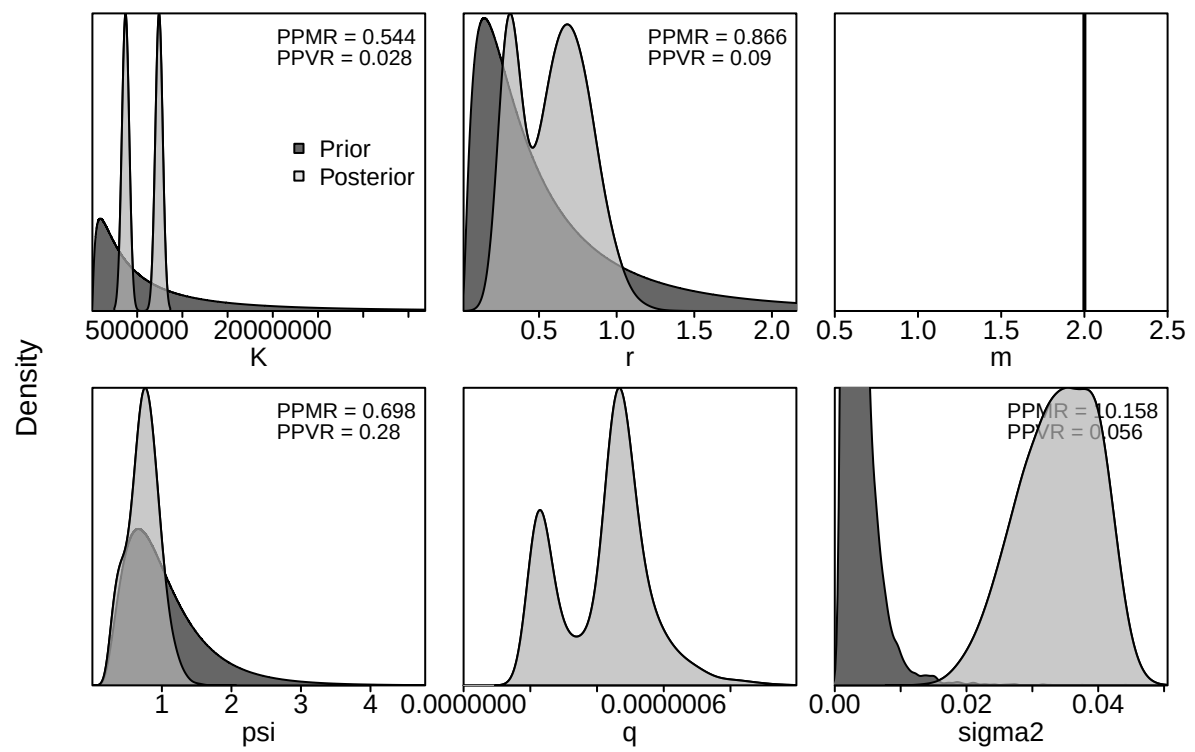


Scenario B

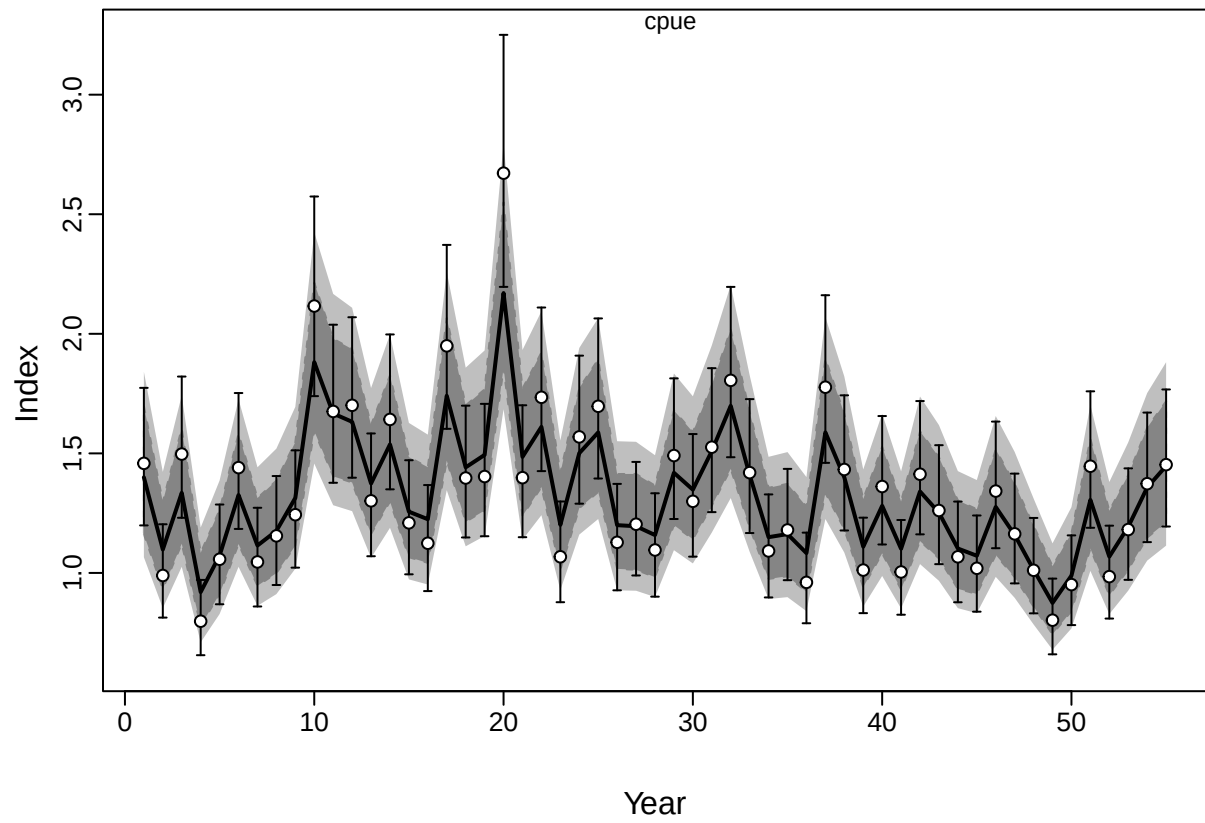
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## ><> jbplot_catcherror() <><
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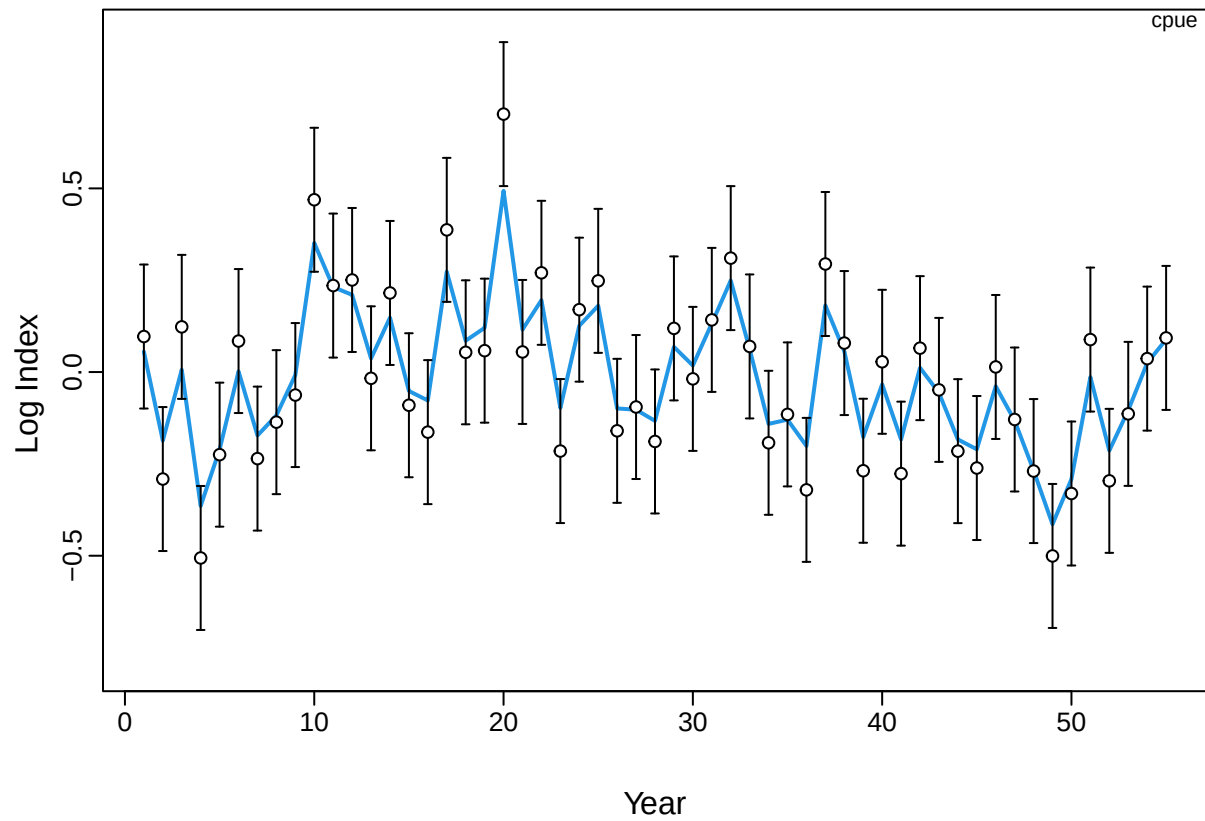
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## <> jplot_ppist() - prior and posterior distributions <>
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## ><> jplot_cpue() - fits to CPUE <><
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## ><> jplot_cpue() - fits to CPUE <><
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##
## ><> jplot_trj() - B trajectory <><

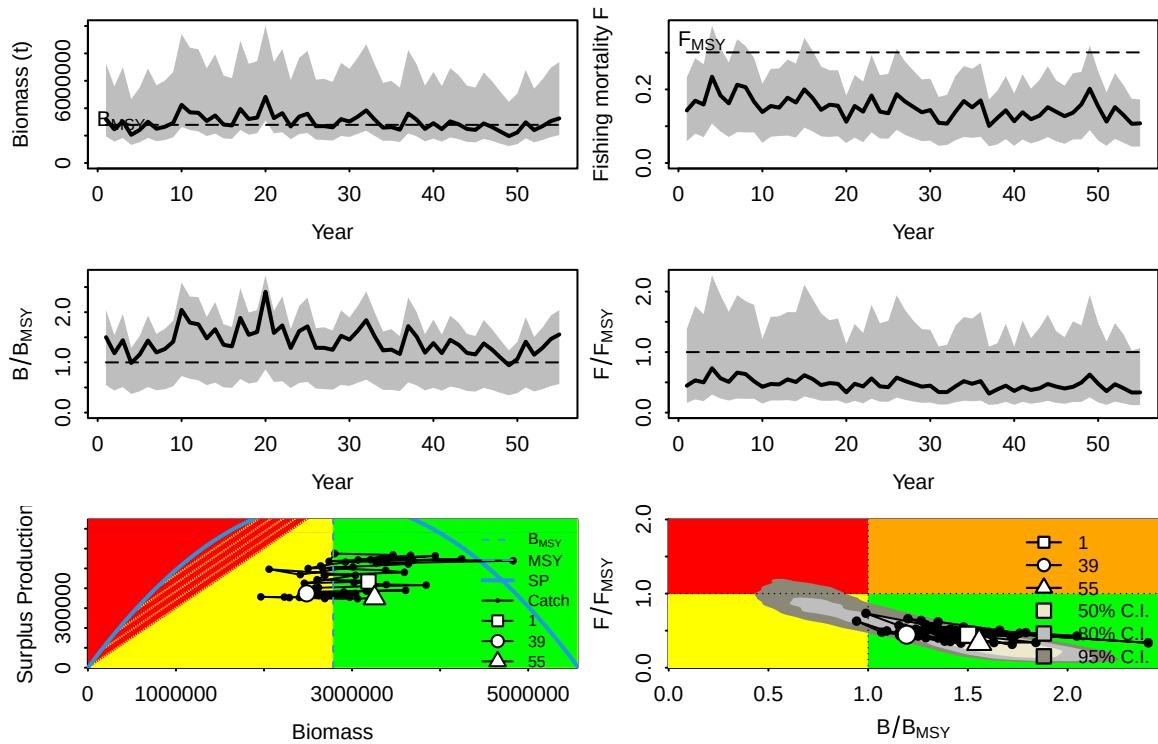
##
## ><> jplot_trj() - F trajectory <><

##
## ><> jplot_trj() - BBmsy trajectory <><

##
## ><> jplot_trj() - FFmsy trajectory <><

##
## ><> jplot_spphase() - JABBA Surplus Production Phase Plot <><

##
## ><> jplot_kobe() - Stock Status Plot <><
```



```

model1 <- bind_rows(MSY_summary_scenA_model1, MSY_summary_scenB_model1)

model1 <- model1 %>%
  mutate(
    bias = meanMSY - true_msy,
    lwr_bias = lwrMSY - true_msy,
    upr_bias = uprMSY - true_msy,
    rel_bias = (meanMSY - true_msy) / true_msy,
    covered = lwrMSY <= true_msy & uprMSY >= true_msy,
    ci_width = uprMSY - lwrMSY)

#MSY
ggplot(aes(x = model, y = meanMSY), data = model1) +
  geom_hline(yintercept = 656250, linetype = "dashed", color = "grey50", alpha = 0.9) +
  geom_linerange(aes(ymin = lwrMSY, ymax=uprMSY), linewidth = 1.2) +
  geom_point(aes(), size = 4) +
  theme_bw() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = ""
  ) +
  labs(
    x = "Model",
    y = "Estimated MSY (kg/yr)",
    title = "Estimated MSY compared to True MSY - Uninformative Priors"
  ) +

```

```
theme(text = element_text(family = "serif", size = 16),
      axis.title.x = element_text(margin = margin(t = 7), size = 14),
      axis.title.y = element_text(margin = margin(r = 10)))
```

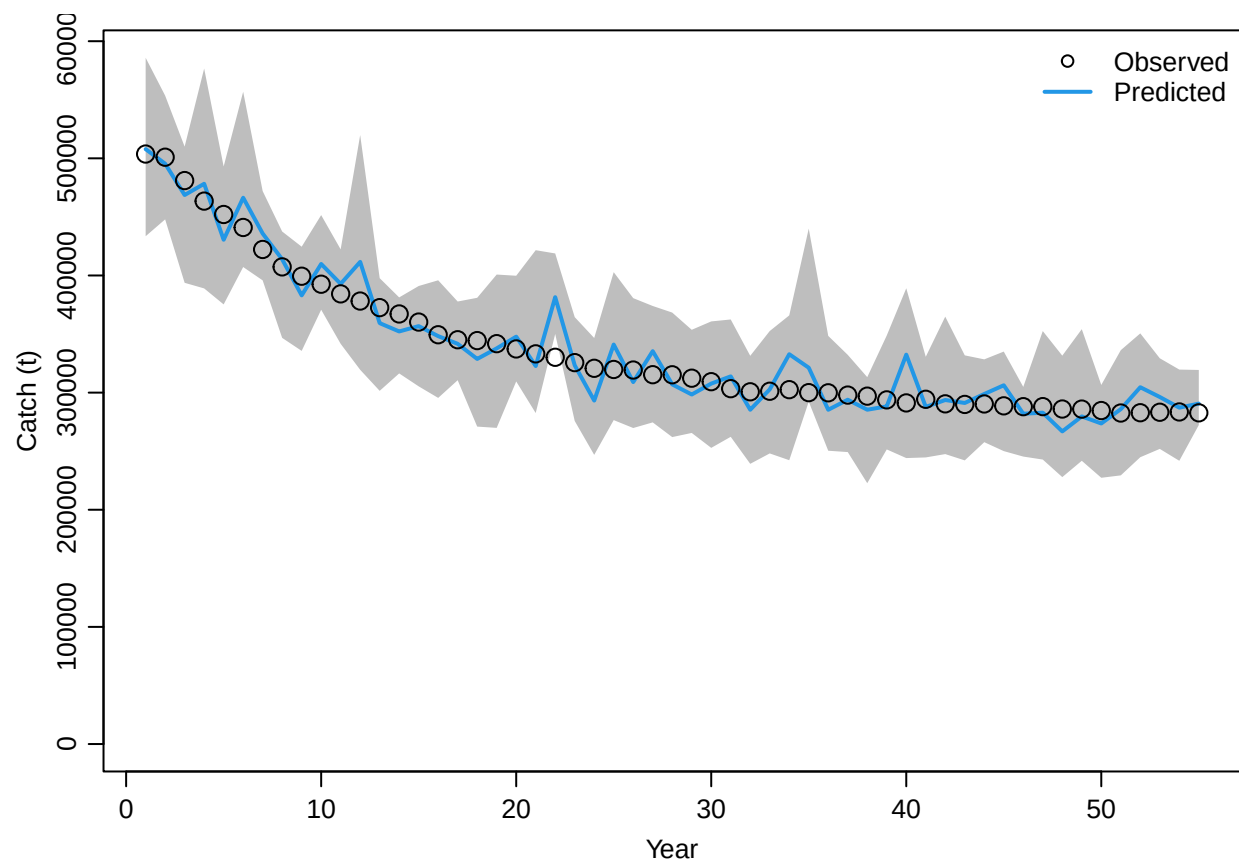
Estimated MSY compared to True MSY – Uninformative



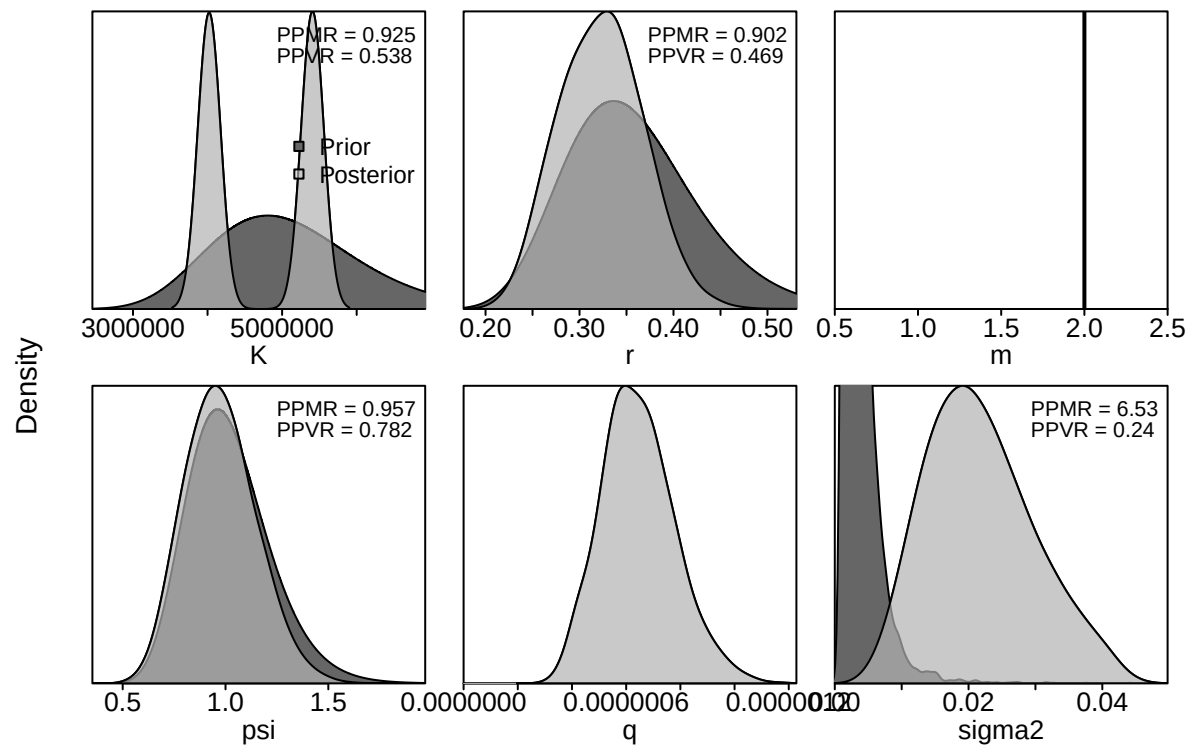
Model 2: Standard Surplus Production + Informative Priors

Scenario A

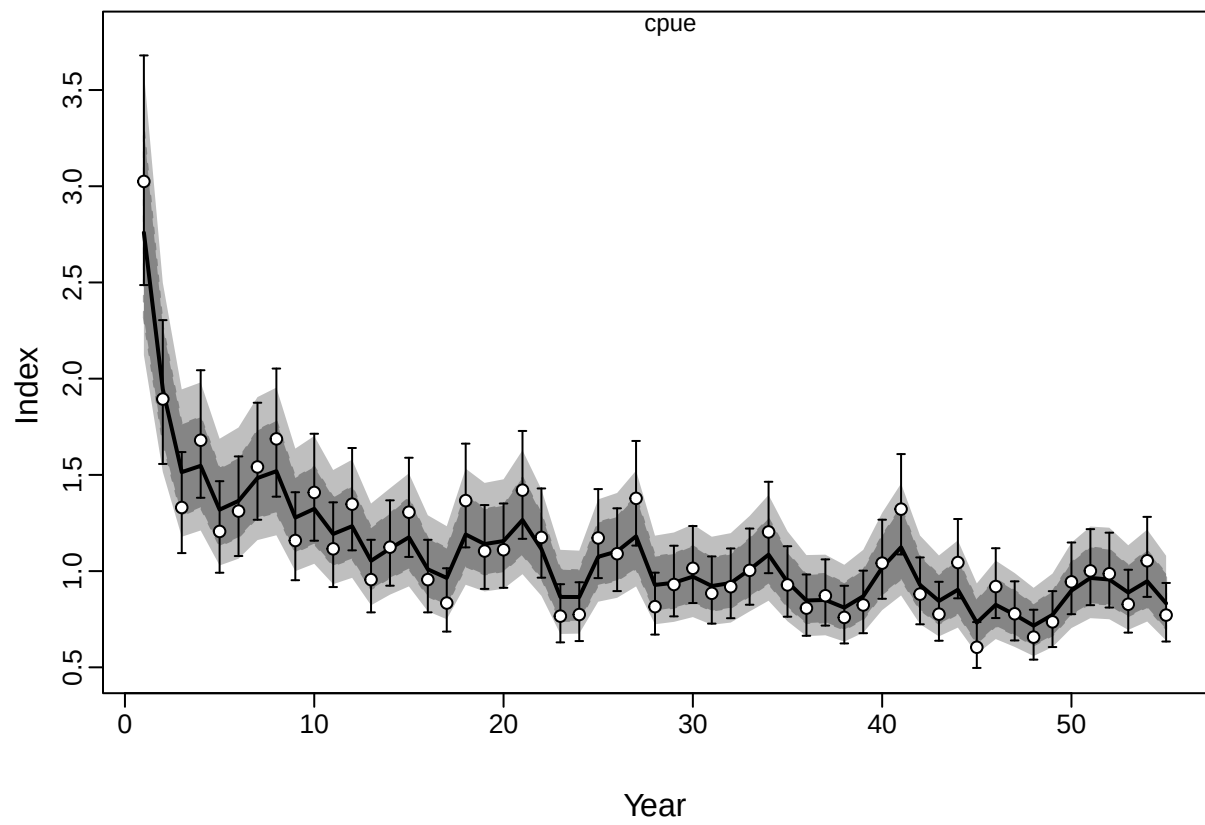
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##
## ><> jplot_catcherror() <><
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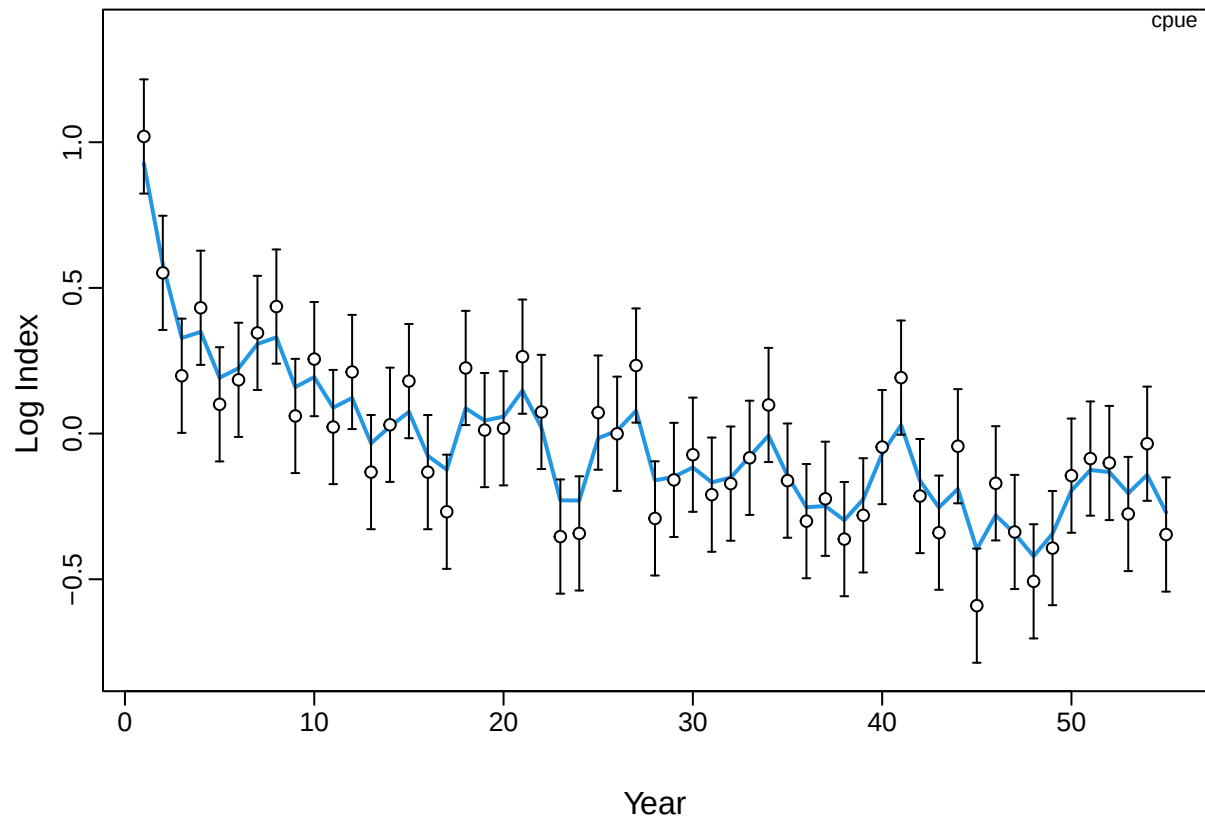
```
##
## <> jbplot_ppist() - prior and posterior distributions <>
```



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##
## ><> jplot_cpue() - fits to CPUE <><
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##
## ><> jplot_cpue() - fits to CPUE <><
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##
## ><> jplot_trj() - B trajectory <><

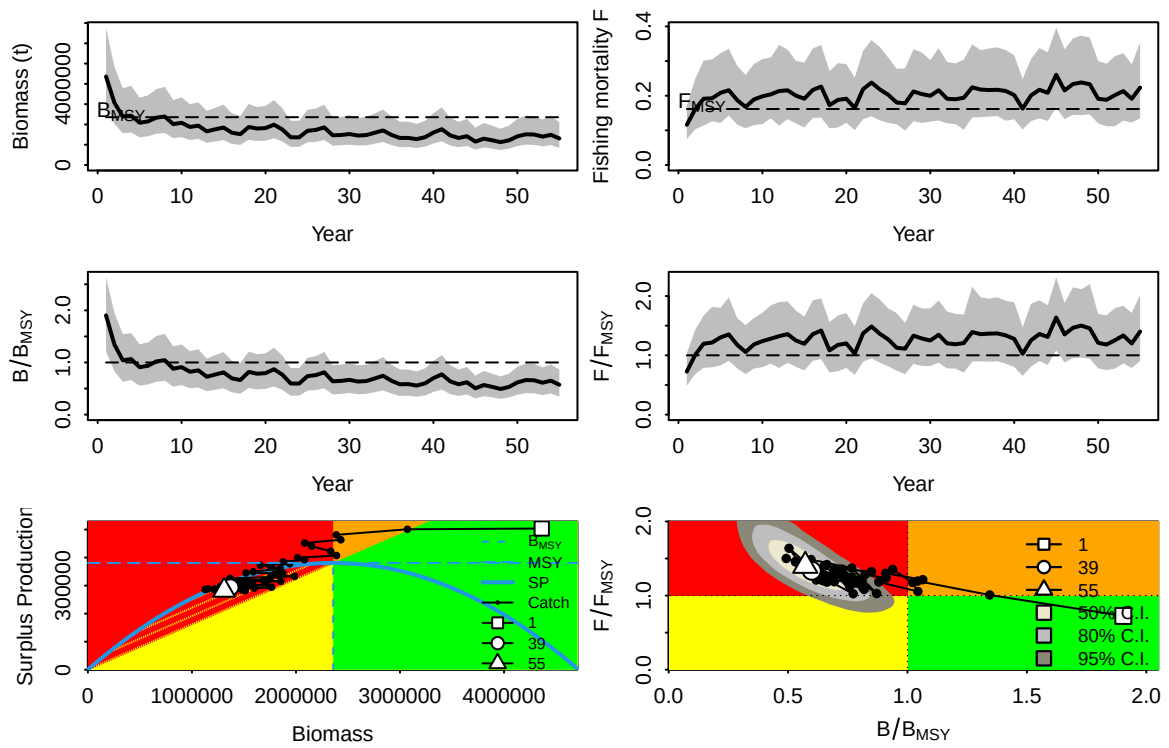
##
## ><> jplot_trj() - F trajectory <><

##
## ><> jplot_trj() - BBmsy trajectory <><

##
## ><> jplot_trj() - FFmsy trajectory <><

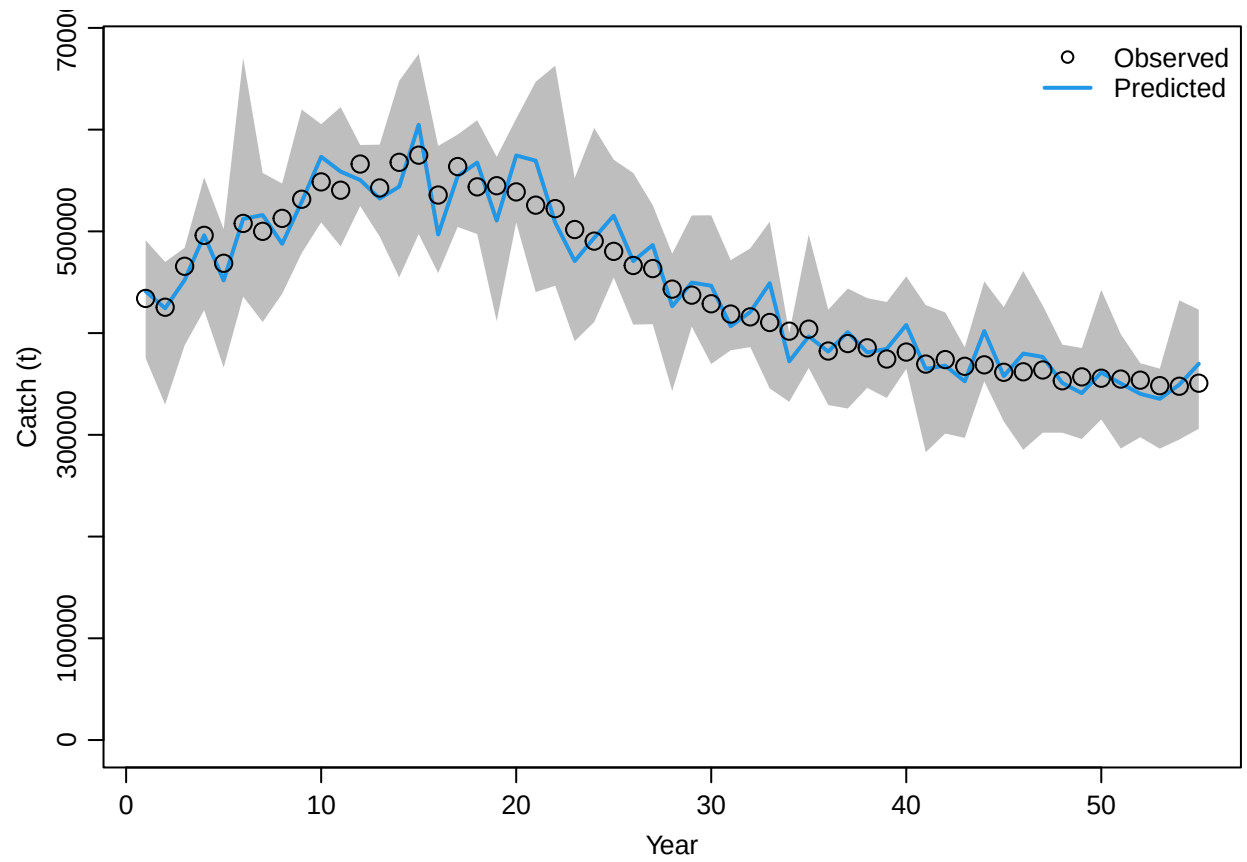
##
## ><> jplot_spphase() - JABBA Surplus Production Phase Plot <><

##
## ><> jplot_kobe() - Stock Status Plot <><
```

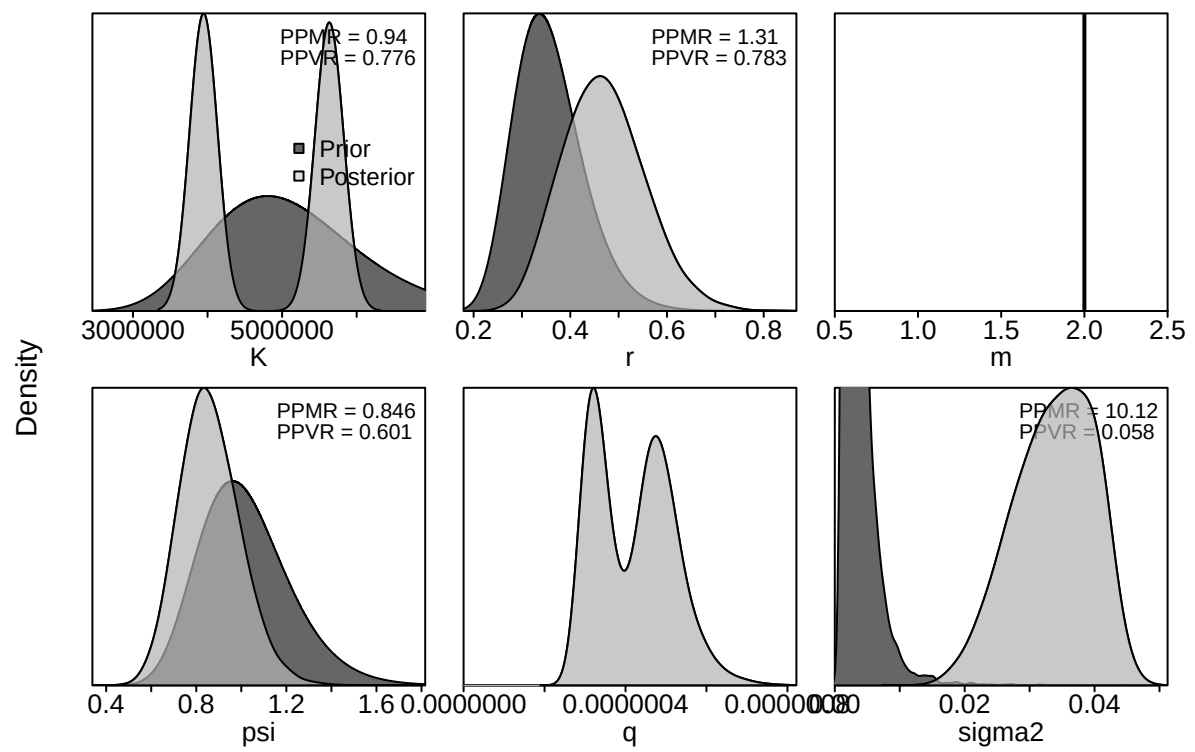


Scenario B

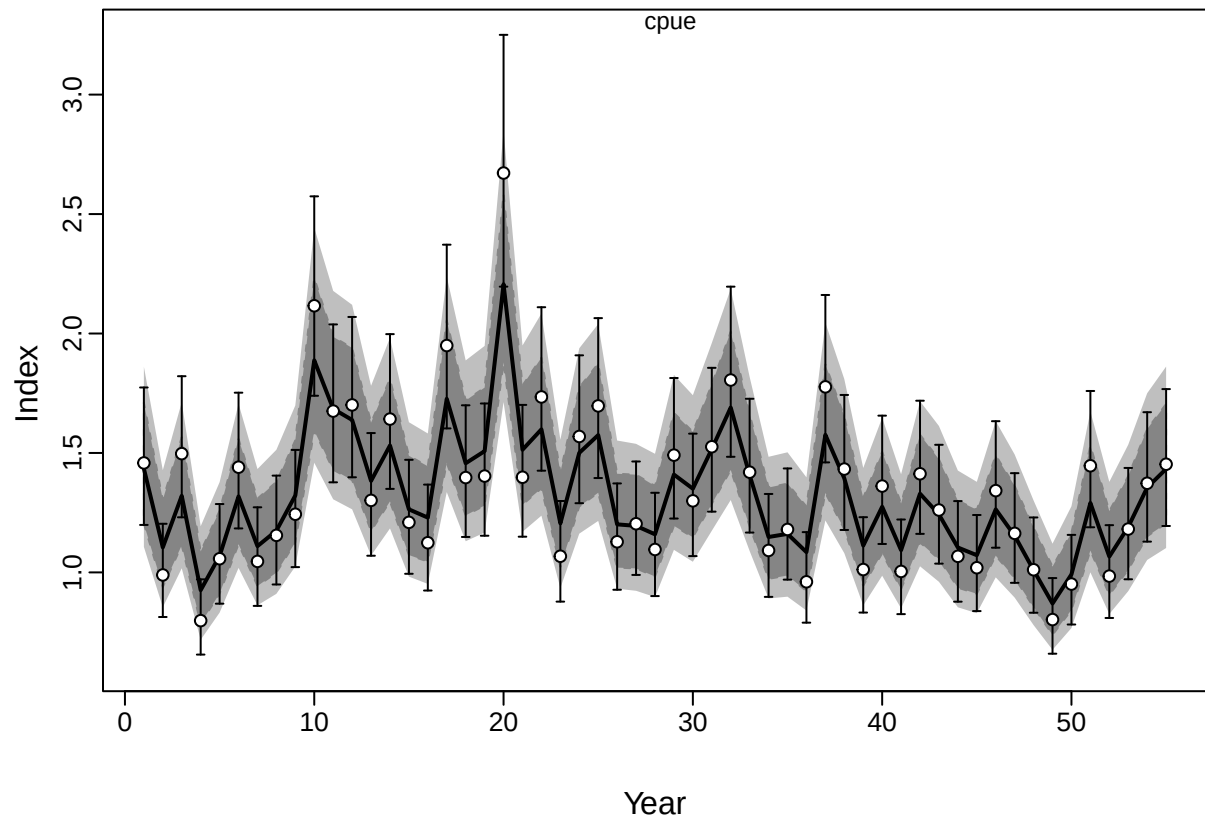
```
##
## ><> jbplot_catcherror() <><
```



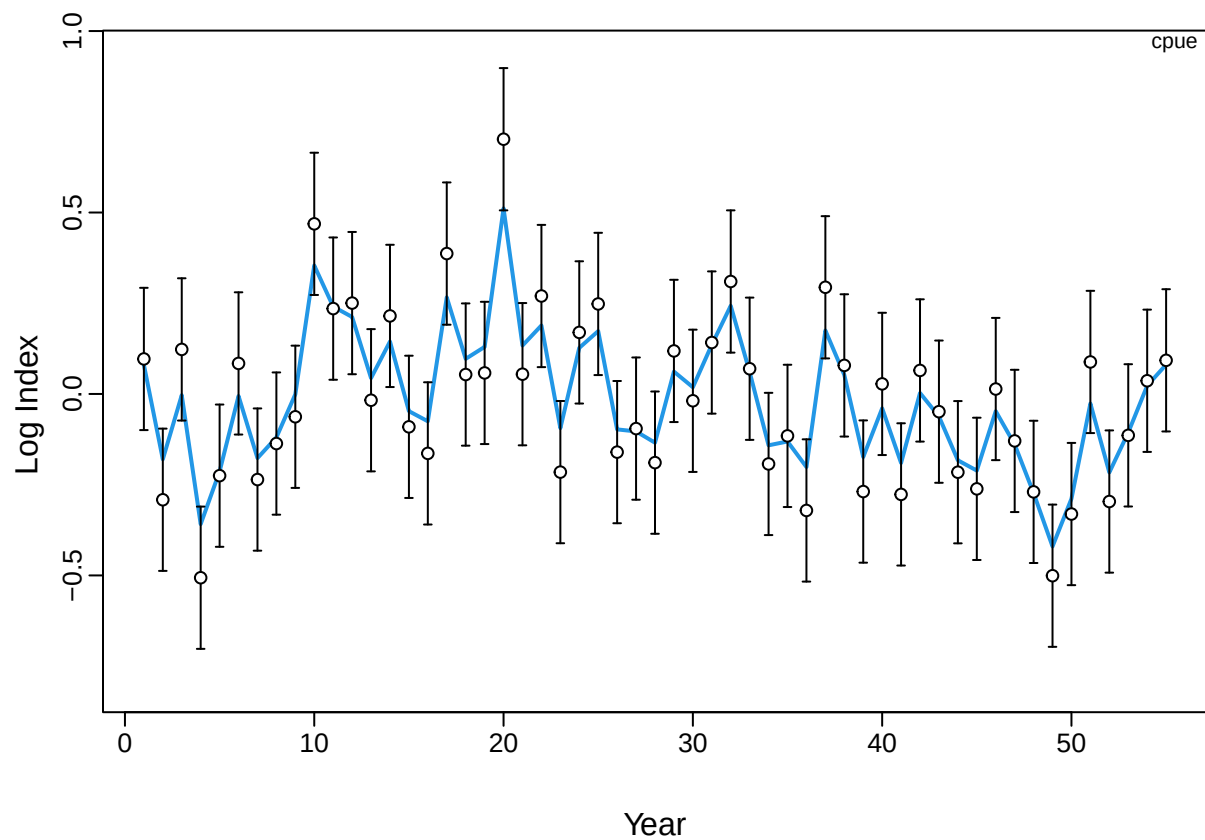
```
##
## >< jplot_ppist() - prior and posterior distributions <><
```



```
##
## ><> jplot_cpue() - fits to CPUE <><
```



```
##
## ><> jplot_cpue() - fits to CPUE <><
```

```
##
## ><> jplot_trj() - B trajectory <><

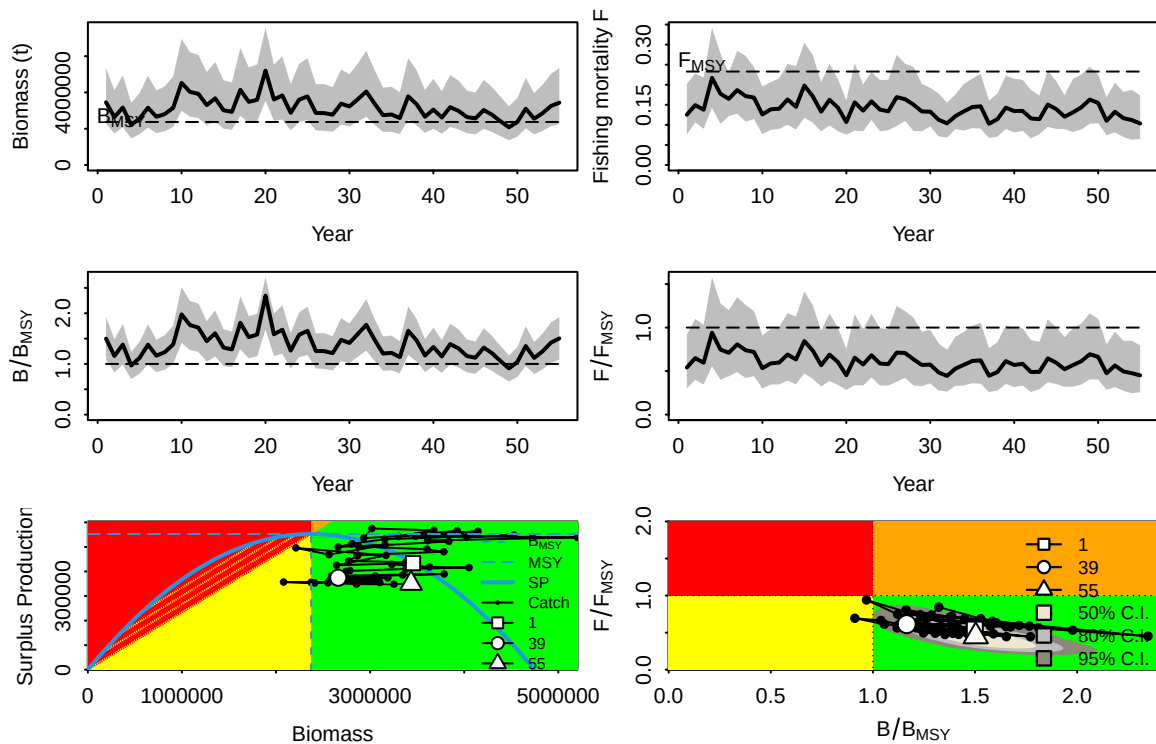
##
## ><> jplot_trj() - F trajectory <><

##
## ><> jplot_trj() - BBmsy trajectory <><

##
## ><> jplot_trj() - FFmsy trajectory <><

##
## ><> jplot_spphase() - JABBA Surplus Production Phase Plot <><

##
## ><> jplot_kobe() - Stock Status Plot <><
```



```

model2 <- bind_rows(MSY_summary_scenA_model2, MSY_summary_scenB_model2)

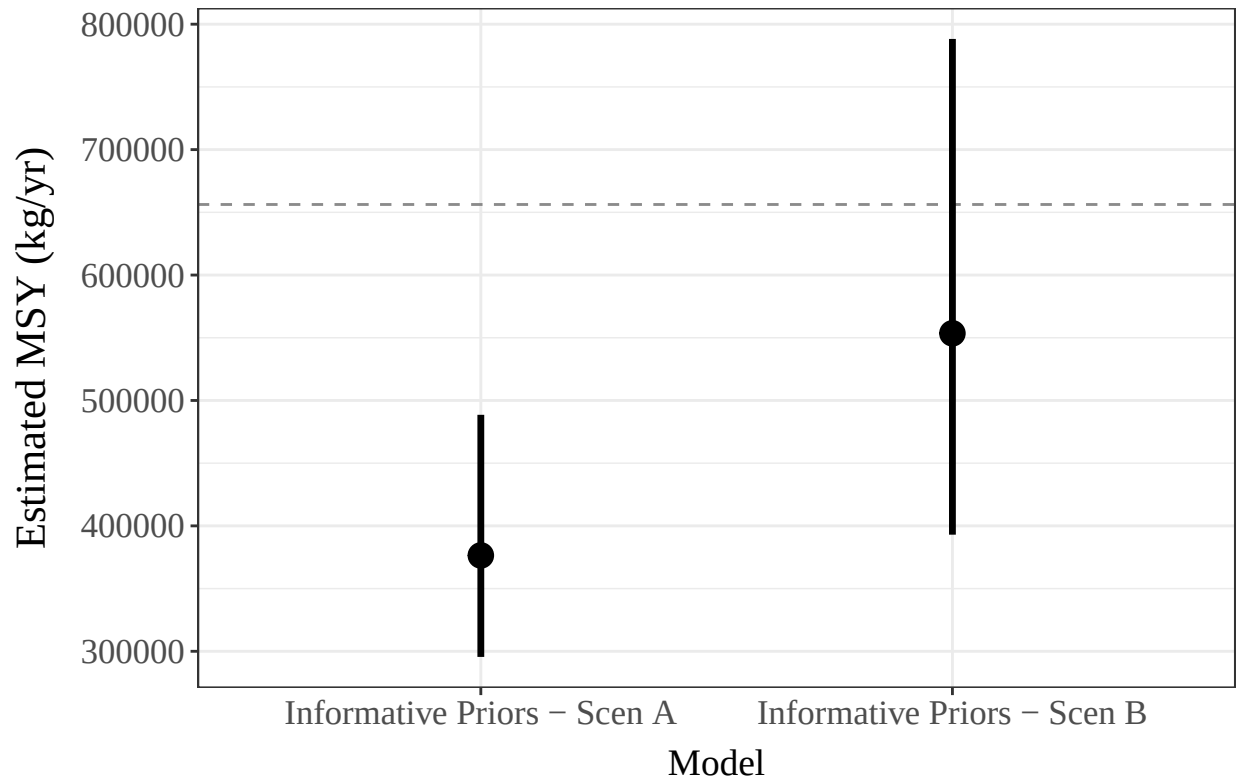
model2 <- model2 %>%
  mutate(
    bias = meanMSY - true_msy,
    lwr_bias = lwrMSY - true_msy,
    upr_bias = uprMSY - true_msy,
    rel_bias = (meanMSY - true_msy) / true_msy,
    covered = lwrMSY <= true_msy & uprMSY >= true_msy,
    ci_width = uprMSY - lwrMSY)

#MSY
ggplot(aes(x = model, y = meanMSY), data = model2) +
  geom_hline(yintercept = 656250, linetype = "dashed", color = "grey50", alpha = 0.9) +
  geom_linerange(aes(ymin = lwrMSY, ymax=uprMSY), linewidth = 1.2) +
  geom_point(aes(), size = 4) +
  theme_bw() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = ""
  ) +
  labs(
    x = "Model",
    y = "Estimated MSY (kg/yr)",
    title = "Estimated MSY compared to True MSY - Informative Priors"
  ) +

```

```
theme(text = element_text(family = "serif", size = 16),
      axis.title.x = element_text(margin = margin(t = 7), size = 14),
      axis.title.y = element_text(margin = margin(r = 10)))
```

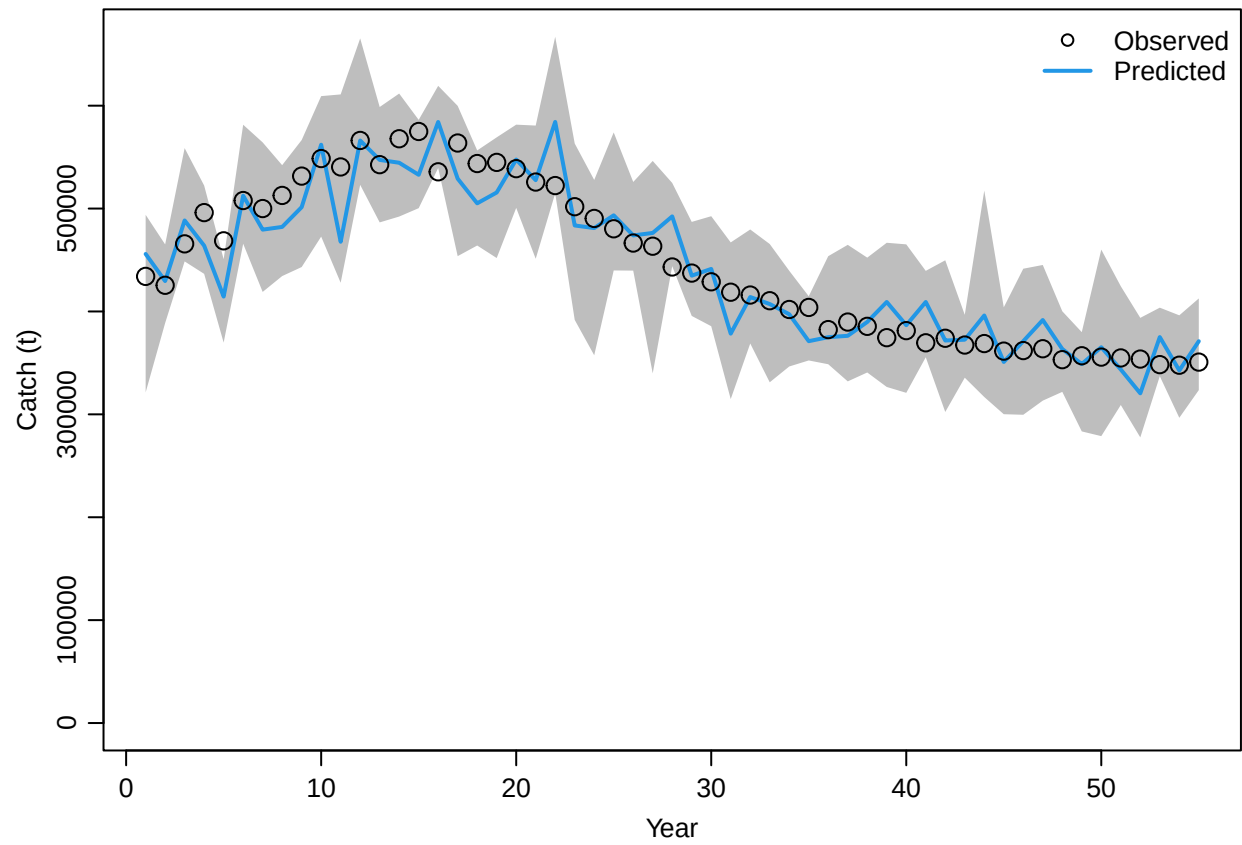
Estimated MSY compared to True MSY – Informative



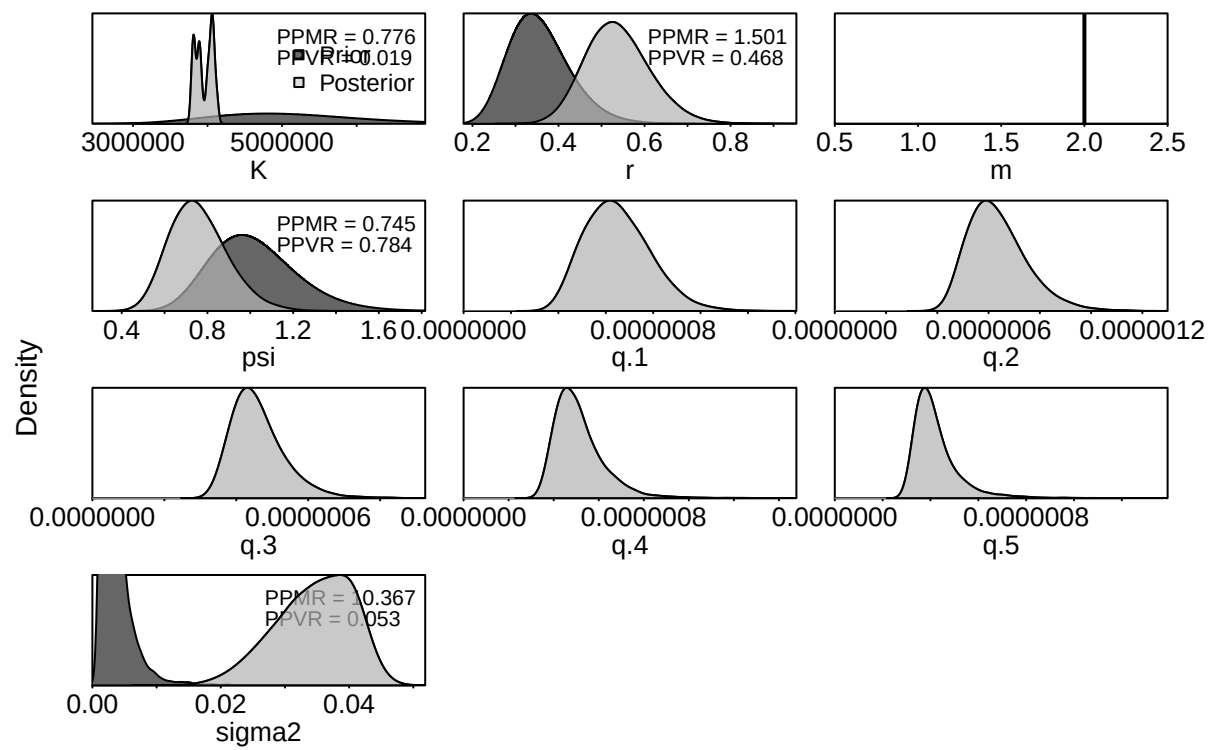
Model 3: Flexible Catchability

This model uses the `sets.q` function within JABBA to assign catchability to different CPUE indices. This subsets catchability into five different time periods.

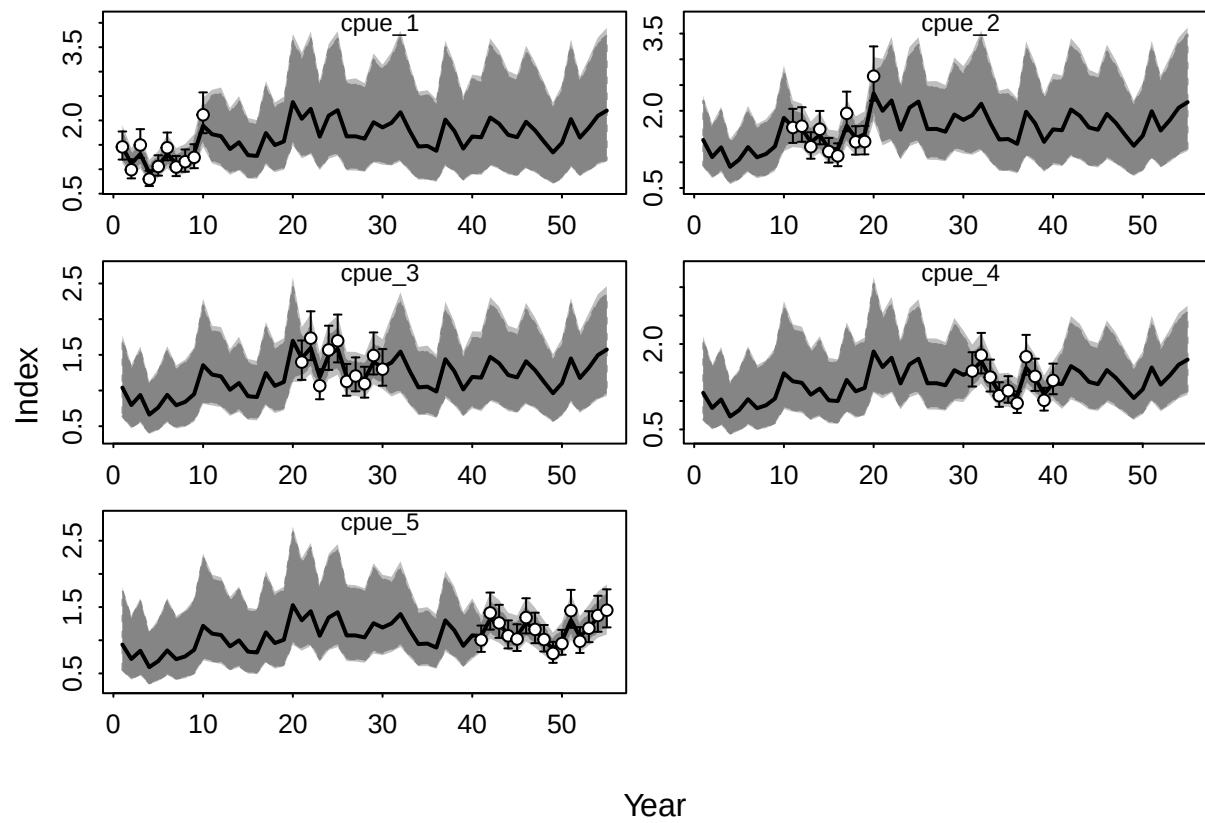
```
##
## ><> jplot_catcherror() <><
```



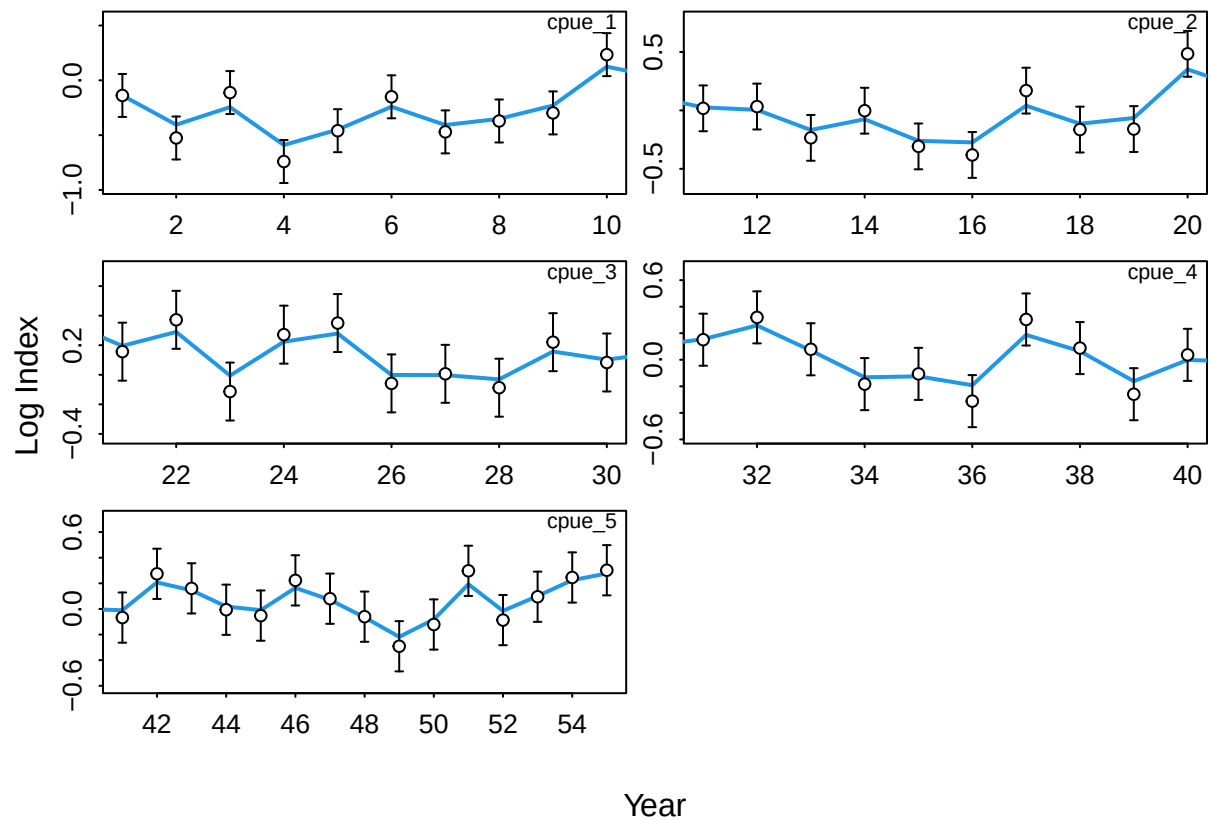
```
##
## >< jplot_ppist() - prior and posterior distributions <><
```



```
##
## ><> jplot_cpue() - fits to CPUE <><
```



```
##
## ><> jplot_cpue() - fits to CPUE <><
```



```
##
## ><> jplot_trj() - B trajectory <><

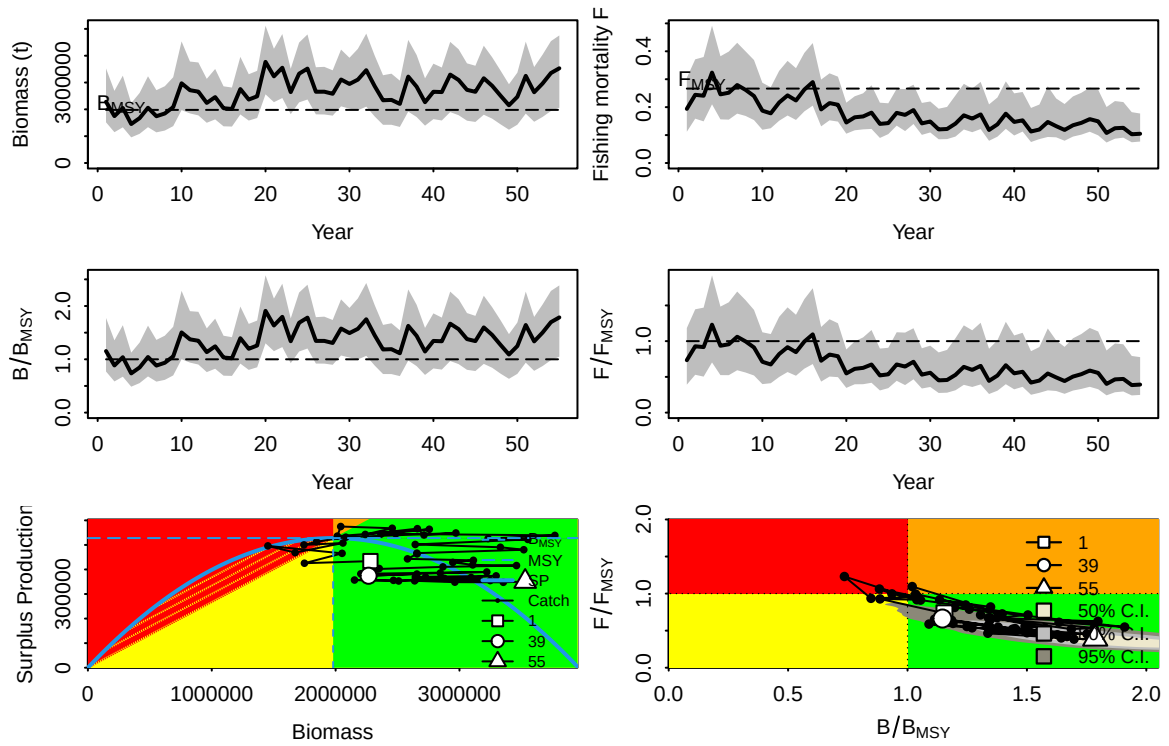
##
## ><> jplot_trj() - F trajectory <><

##
## ><> jplot_trj() - BBmsy trajectory <><

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## ><> jplot_trj() - FFmsy trajectory <><

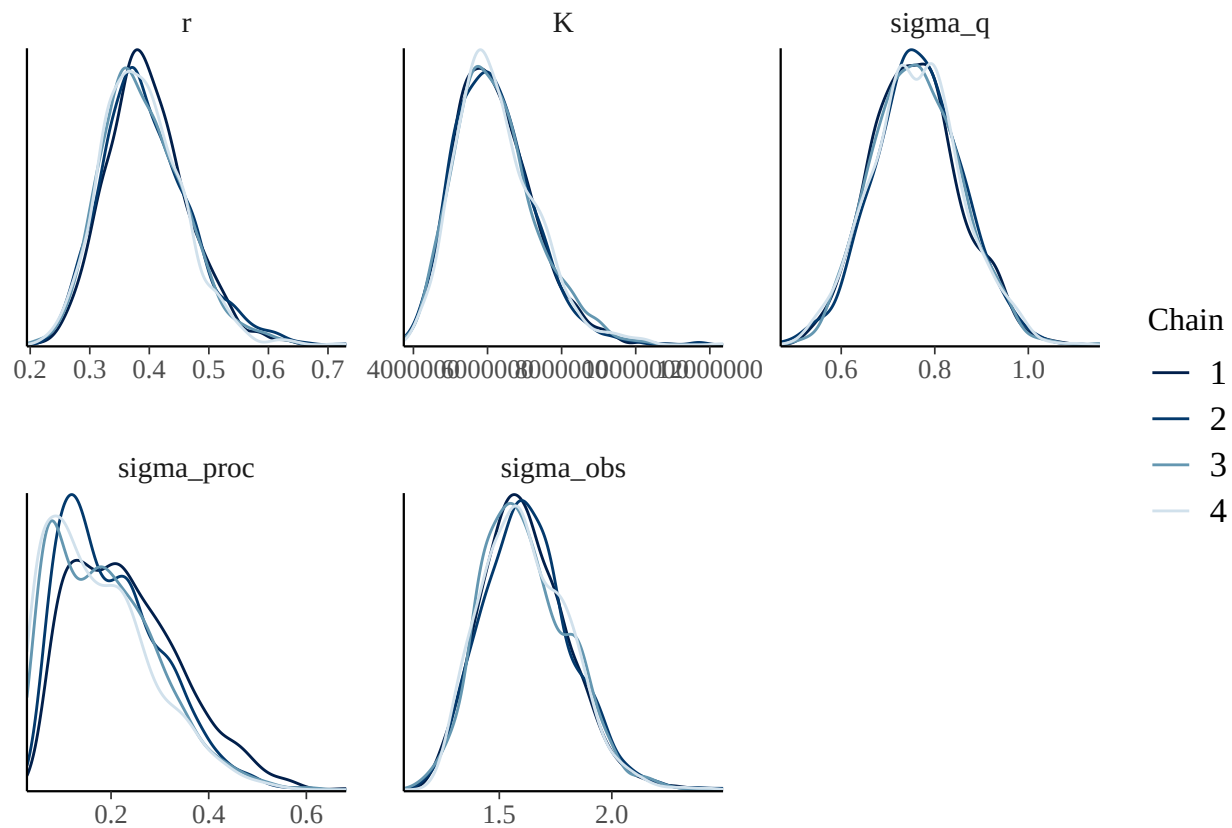
##
## ><> jplot_spphase() - JABBA Surplus Production Phase Plot <><

##
## ><> jplot_kobe() - Stock Status Plot <><
```



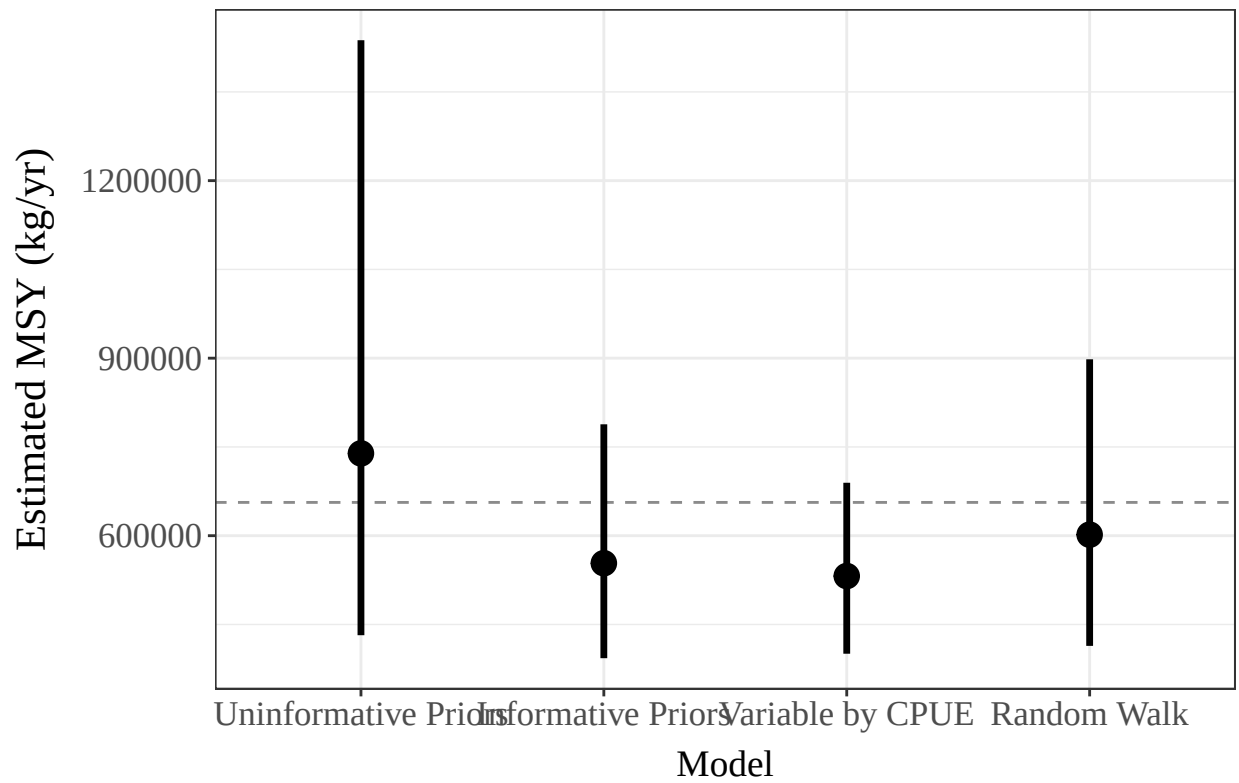
Model 4:

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
r	0.39	0.39	0.07	0.07	0.29	0.51	1.00	2313.51	3033.61
K	6201283.61	6070248.65	1080061.36	1018945.46	4661553.19	8171327.21	1.00	1497.96	2615.04
sigma_q	0.76	0.76	0.09	0.09	0.61	0.92	1.00	1371.53	2163.12
sigma_proc	0.20	0.19	0.11	0.12	0.06	0.40	1.05	74.65	87.81
sigma_obs	1.62	1.60	0.19	0.20	1.33	1.95	1.00	2855.85	3389.00

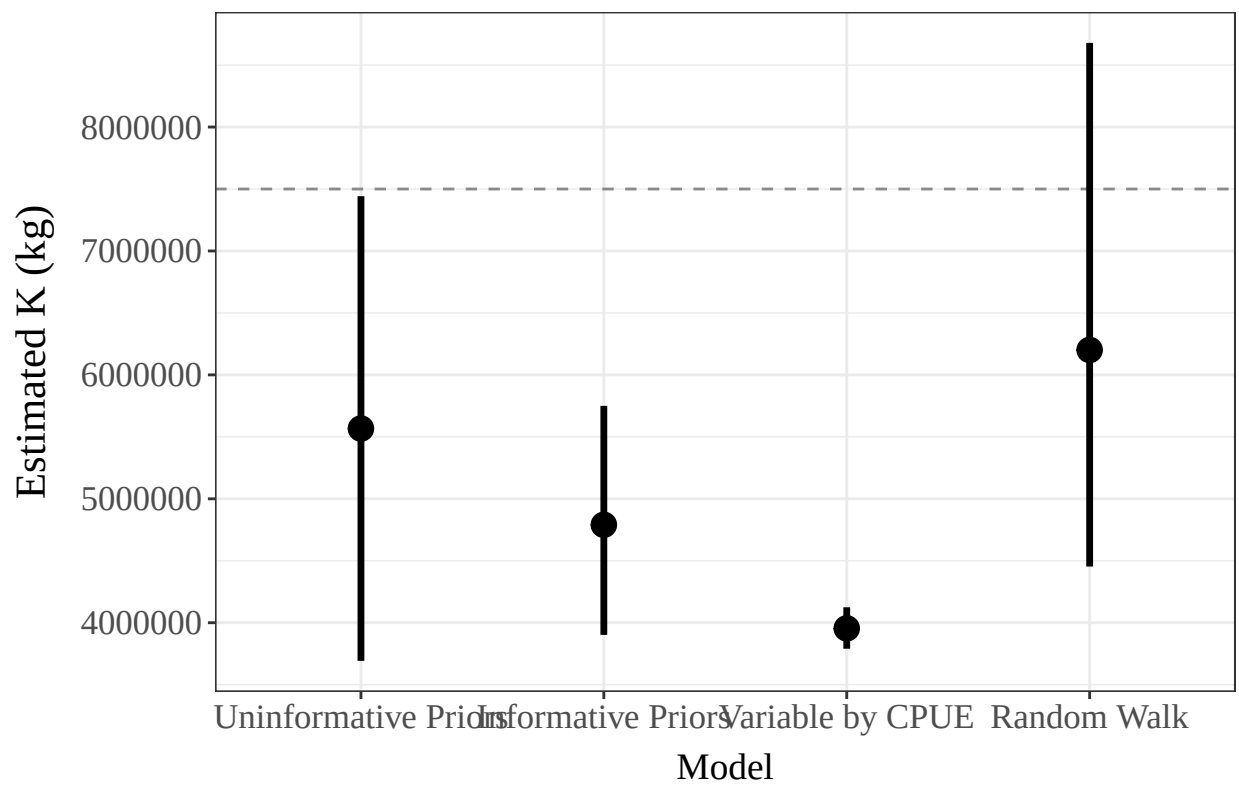


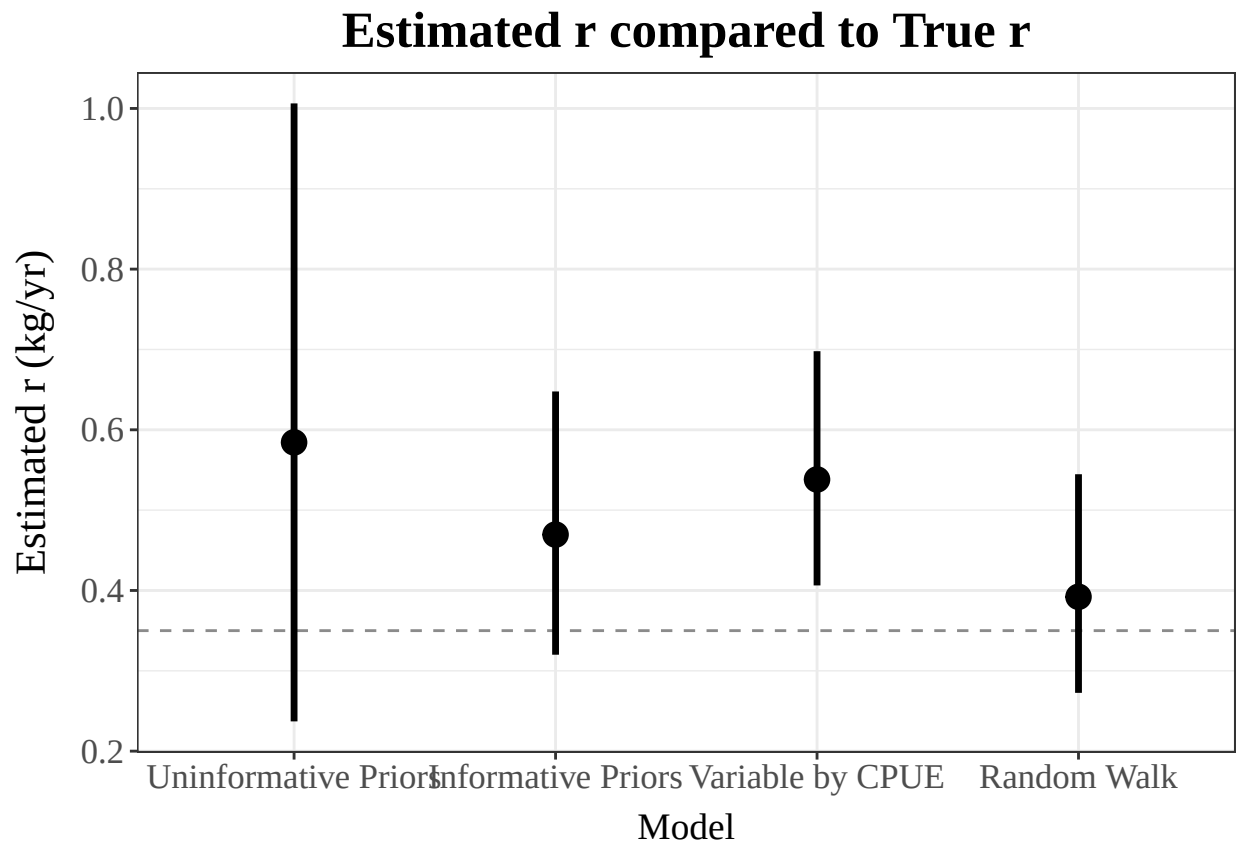
Model Evaluation

Estimated MSY compared to True MSY



Estimated K compared to True K





Each of the models has a confidence interval that contains the true MSY.